Contents

1 Introduction **3**

2 Viscoelastic Fluid Mechanics and the Four-Roll Mill **4**
2.1 General Conservation Laws 5
2.1.1 Mass Balance 5
2.1.2 Momentum Balance 5
2.2 Physical Assumptions 6
2.2.1 Stream Function Formulation 7
2.3 Constitutive Modeling 8
2.3.1 Upper-Convected Time Derivative 9
2.3.2 Oldroyd-B Constitutive Model 10
2.3.3 Giesekus Model 11
2.3.4 Linear Phan-Thien-Tanner (PTT) Model 12
2.4 Dimensionless Governing Equations 12
2.4.1 Reference Scales and Dimensionless Parameters 13
2.4.2 Dimensionless Governing System 14
2.5 Four-Roll Mill 15
2.5.1 Flow Index Parameter 15
2.5.2 Stagnation Point 17
2.5.3 Geometric Setup 18
2.5.4 Operating Parameters 18
2.5.5 Boundary Conditions 18

3 Physics-Informed Neural Networks in Complex Rheology **21**
3.1 Fundamentals of Deep Learning 22
3.1.1 The Artificial Neuron and Non-Linear Activation Functions 22
3.1.2 Multi-Layer Perceptron Architecture 23
3.1.3 Loss Functions and Optimization Algorithms 25
3.1.4 Backpropagation 28
3.2 Neural Networks: Capabilities and Physical Limitations 28
3.2.1 Key Advantages 28
3.2.2 Limits of Purely Data-Driven Machine Learning 29
3.3 Physics-Informed Neural Network 30
3.3.1 Physics-Constrained Loss Formulation 30
3.3.2 PINN-Specific Features 33

3.4 Foundational Literature Benchmarks 35
3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019) . 35
3.4.2 The ViscoelasticNet Framework 36

4 Methodology and Numerical Results 39
4.1 Problem Formulation 40
4.1.1 Recirculating Cavity Challenges 40
4.1.2 Forward and Inverse Modes 40
4.2 Reference CFD Dataset 41
4.2.1 Finite Element Setup 41
4.2.2 Mesh Convergence Study 43
4.3 Network Architecture and Loss Formulation 43
4.3.1 Decoupled Three-Head Network 43
4.3.2 Weight Initialization 47
4.3.3 Governing Residuals and Scaling 47
4.3.4 Composite Loss Function and Roller Stress Anchor 49
4.4 Training Strategy 52
4.4.1 Two-Stage Physical Decoupling 52
4.4.2 Dual-Precision Optimization 54
4.4.3 Transfer Learning Protocol 54
4.5 Direct Benchmark: Latent Stress and Pressure Reconstruction . . 55
4.5.1 Velocity and Stress Reconstruction 56
4.5.2 Pressure Recovery and Gauge Calibration 57
4.6 Inverse Parameter Discovery 59
4.6.1 Oldroyd-B and Giesekus Benchmarks 59
4.6.2 Transfer Learning Performance 62
4.6.3 Collocation Density Sensitivity 62
4.7 Identifiability Boundaries and Invariance Principles 63
4.7.1 Elastic Shear Modulus Invariance 63
4.7.2 Creeping Flow Hydrodynamic Invariance 66
4.7.3 Shear Degeneracy in Linear PTT 66

5 Conclusions and Future Perspectives 70
5.1 Summary of Key Findings and Contributions 70
5.2 Operational Limitations 70
5.3 Future Perspectives and Research Directions 70

Chapter 1

Introduction

PLACEHOLDER

Chapter 2

Viscoelastic Fluid Mechanics and the Four-Roll Mill

The hydrodynamics of dilute and semi-dilute polymer solutions fundamentally departs from classical Newtonian fluid behavior. Macromolecular coils dissolved in a solvent undergo continuous conformational changes under applied flow fields, stretching along extensional trajectories and storing elastic energy with a characteristic relaxation memory. Consequently, the local stress state in polymeric fluids depends not merely on the instantaneous rate of deformation, but integrates the entire kinematic history experienced by fluid elements. Accurately modeling such transport phenomena requires a continuum mechanics formulation where universal conservation laws are coupled with non-linear, frame-indifferent constitutive equations.

In this chapter, we develop the continuum framework governing viscoelastic flows within the four-roll mill geometry. Starting from the foundational balance laws of mass and linear momentum, we introduce the kinematic reductions characteristic of confined, low-inertia polymeric flows. We then establish the constitutive formulation of viscoelasticity through the Upper-Convected Time Derivative, analyzing both the classical Oldroyd-B model, including the physical roots of its

extensional stress singularity, and non-linear stress-saturating closures such as the Giesekus and Phan-Thien-Tanner (PTT) formulations. Finally, we cast the governing system into dimensionless form using a viscous stress scale, detail the geometry and hyperbolic stagnation kinematics of the four-roll mill, and prescribe the boundary constraints that establish the boundary value problem.

2.1 General Conservation Laws

From a macroscopic continuum perspective, fluid motion is governed by the universal conservation laws of mass and linear momentum [1].

2.1.1 Mass Balance

For a fluid possessing local density $\rho(\boldsymbol{x}, t)$, the conservation of mass over an arbitrary control volume is expressed by the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0. \tag{2.1}$$

2.1.2 Momentum Balance

Applying Newton's second law of motion to an infinitesimal fluid element yields the local momentum balance:

$$\rho \frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}, \tag{2.2}$$

where $\boldsymbol{g}$ represents external body force accelerations and $\boldsymbol{\sigma}$ denotes the symmetric total stress tensor. The left-hand side operator $\frac{\mathrm{D}}{\mathrm{D}t}$ is the substantial (or material) derivative, defined as:

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} \equiv \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}. \tag{2.3}$$

In the Eulerian framework, the substantial derivative expresses the total rate of velocity change experienced by a moving fluid parcel, including the time derivative $\frac{\partial \boldsymbol{u}}{\partial t}$ at a fixed spatial position and the convective part $(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$. Expanding Equation (2.2):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}. \tag{2.4}$$

In order to separate isotropic hydrostatic effects from deviatoric shearing and stretching deformations, the total stress tensor $\boldsymbol{\sigma}$ is split into an isotropic pressure contribution and an extra-stress tensor $\boldsymbol{T}$:

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + \boldsymbol{T}, \tag{2.5}$$

where p is the isotropic hydrodynamic pressure, $\boldsymbol{I}$ is the identity tensor, and $\boldsymbol{T}$ represents all viscous and viscoelastic stresses. Substituting Equation (2.5) into Equation (2.4):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{T} + \rho \boldsymbol{g}. \tag{2.6}$$

2.2 Physical Assumptions

The general conservation equations can be simplified by introducing the scale-appropriate physical hypotheses of the four-roll mill:

1. **Incompressibility**: Under moderate operating pressures and low characteristic velocities, density variations are negligible ($\rho = \text{const}$). The continuity equation (2.1) simplifies to the divergence-free condition:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{2.7}$$

2. **Steady-state regime**: We focus on the fully established operating regime,

setting all explicit time derivatives to zero:

$$\frac{\partial(\cdot)}{\partial t} = \mathbf{0}. \tag{2.8}$$

3. **Absence of body forces**: The four-roll mill operates in a horizontal cross-sectional plane (x, y). Because gravity acts along the vertical cylinder axis (z), perpendicular to the flow plane, external body forces vanish: $\boldsymbol{g} = \mathbf{0}$.

4. **Isothermal operating conditions**: Because velocities are low and viscous heating is negligible, the temperature field remains uniform. Fluid transport properties, including viscosities and relaxation times, are therefore treated as constants.

5. **Two-dimensional planar kinematics**: Given the large aspect ratio of the cylindrical rollers along the z-axis, three-dimensional end effects are negligible in the transverse mid-plane. The velocity field is modeled as strictly planar:

$$\boldsymbol{u} = \begin{pmatrix} u(x,y) \\ v(x,y) \\ 0 \end{pmatrix}. \tag{2.9}$$

2.2.1 Stream Function Formulation

In two-dimensional incompressible flow, the solenoidal constraint $\nabla \cdot \boldsymbol{u} = 0$ can be satisfied directly by defining a scalar stream function $\psi(x, y)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{2.10}$$

Because mixed partial derivatives commute ($\frac{\partial^2\psi}{\partial x\partial y} = \frac{\partial^2\psi}{\partial y\partial x}$), the velocity divergence vanishes identically:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2\psi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial y\partial x} \equiv 0. \tag{2.11}$$

Parameterizing the velocity through ψ satisfies mass conservation everywhere by construction, replacing the two velocity components (u, v) with a single scalar field.

Under these physical assumptions and kinematically reduced velocity representation, the steady balance of linear momentum simplifies to:

$$\rho(\boldsymbol{u} \cdot \nabla\boldsymbol{u}) = -\nabla p + \nabla \cdot \boldsymbol{T}. \tag{2.12}$$

2.3 Constitutive Modeling

In two dimensions, the momentum balance (2.12) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, while mass and momentum conservation provide only three equations. A constitutive equation for the extra-stress tensor $\boldsymbol{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, we split the extra-stress tensor $\boldsymbol{T}$ into a Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\boldsymbol{T} = 2\mu_s\boldsymbol{D} + \boldsymbol{\tau}, \tag{2.13}$$

where μ_s is the dynamic viscosity of the Newtonian solvent, and $\boldsymbol{D}$ is the symmetric rate-of-strain tensor:

$$\boldsymbol{D} = \frac{1}{2}\left[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\right]. \tag{2.14}$$

Taking the divergence of the solvent stress and leveraging incompressibility ($\nabla \cdot \boldsymbol{u} = 0$):

$$\nabla \cdot (2\mu_s \boldsymbol{D}) = \mu_s \nabla \cdot \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right) = \mu_s \left(\nabla^2 \boldsymbol{u} + \nabla(\nabla \cdot \boldsymbol{u})\right) = \mu_s \nabla^2 \boldsymbol{u}. \quad (2.15)$$

Substituting Equation (2.15) into Equation (2.12) yields the steady momentum equation:

$$\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) = -\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau}. \quad (2.16)$$

2.3.1 Upper-Convected Time Derivative

Polymer chains suspended in a solvent stretch, orient, and relax over finite timescales. Because fluid parcels translate, rotate, and deform as they move through the domain, constitutive relations require a rate of stress that is invariant under changes of reference frame (material frame-indifference). A standard time derivative evaluated at a fixed point would interpret rigid-body rotation or parcel stretching as a change in stress, even if the internal state of the polymer remained unaltered.

To measure true rheological evolution, the time rate of change must be evaluated in a local coordinate system that deforms affinely with the fluid element. This objective tensor rate is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T, \quad (2.17)$$

where $(\nabla \boldsymbol{u})_{ij} = \partial u_i / \partial x_j$. For steady flows ($\partial \boldsymbol{\tau}/\partial t = \boldsymbol{0}$), the operator combines advective stress transport along streamlines, $\boldsymbol{u} \cdot \nabla \boldsymbol{\tau}$, with kinematic stretching and rotation induced by local velocity gradients, $-(\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T$.

2.3.2 Oldroyd-B Constitutive Model

The Oldroyd-B model [4] provides the standard starting point for viscoelastic fluid modeling. It idealizes macromolecules as non-interacting Hookean dumbbells suspended in a Newtonian solvent [1]. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.18}$$

where λ is the relaxation time and μ_p is the polymeric viscosity. The total zero-shear dynamic viscosity of the solution is:

$$\mu_0 = \mu_s + \mu_p. \tag{2.19}$$

In steady simple shear flow, Oldroyd-B predicts a constant viscosity $\eta(\dot{\gamma}) = \mu_0$, reproducing the behavior of Boger fluids.

Planar Extensional Behavior

In strong extensional flows, the Oldroyd-B model exhibits an unphysical stress singularity. For a steady planar elongational flow centered at the origin, $u = Ex$ and $v = -Ey$ with extension rate $E > 0$, the tensile normal stress along the stretching axis is:

$$\tau_{xx} = \frac{2\mu_p E}{1 - 2\lambda E}. \tag{2.20}$$

As the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \tag{2.21}$$

the denominator vanishes. Because Hookean springs have infinite extensibility, the polymer chains can stretch without bound, driving both τ_{xx} and the exten-

sional viscosity η_E to infinity.

This stress singularity near $2\lambda E \to 1$ is the constitutive origin of the *High-Weissenberg Number Problem* (HWNP). Near extensional stagnation points, it generates steep stress layers that cause numerical instability in standard mesh-based solvers [3].

2.3.3 Giesekus Model

To overcome the unphysical stress singularity of Oldroyd-B, Giesekus [1, 20] introduced a quadratic non-linear stress contribution derived from configuration-dependent hydrodynamic drag. In semi-dilute solutions, stretching macromolecular chains align parallel to each other, reducing drag along the orientation direction relative to the transverse directions. This anisotropic friction is incorporated by modifying the stress relaxation rate:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}\boldsymbol{\tau}^2 = 2\mu_p \boldsymbol{D}, \tag{2.22}$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and $\boldsymbol{\tau}^2 = \boldsymbol{\tau} \cdot \boldsymbol{\tau}$. When $\alpha = 0$, isotropic drag is recovered and the model reduces identically to Oldroyd-B. For $\alpha > 0$, the quadratic dissipation term $\boldsymbol{\tau}^2$ provides two key rheological enhancements:

1. **Bounded extensional plateau**: Extensional stresses no longer diverge at E_{crit}; instead, η_E reaches an asymptotic finite plateau at high strain rates, completely resolving the constitutive singularity.

2. **Shear-thinning behavior**: The steady shear viscosity $\eta(\dot{\gamma})$ decreases monotonically with increasing shear rate, closely matching experimental polymer melt and concentrated solution data.

2.3.4 Linear Phan-Thien-Tanner (PTT) Model

Derived from the transient network theory of concentrated polymer solutions and melts [1,21], the Phan-Thien-Tanner (PTT) model treats macromolecules as a network of temporary entanglements. Under strong deformation, mechanical tension on the network strands accelerates junction destruction. In its linear formulation, this stress-induced network breakdown is modeled by scaling the linear relaxation term:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.23}$$

where the scalar function $f(\boldsymbol{\tau})$ is defined as:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau}), \tag{2.24}$$

with $\mathrm{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$ (since $\tau_{zz} = 0$ under planar deformation for linear PTT) and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter. When $\varepsilon = 0$, $f(\boldsymbol{\tau}) \equiv 1$ and the linear Oldroyd-B constitutive law is recovered. For $\varepsilon > 0$, high trace stresses accelerate stress relaxation, bounding extensional viscosity growth and preventing unbounded coil stretch near extensional stagnation points.

2.4 Dimensionless Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

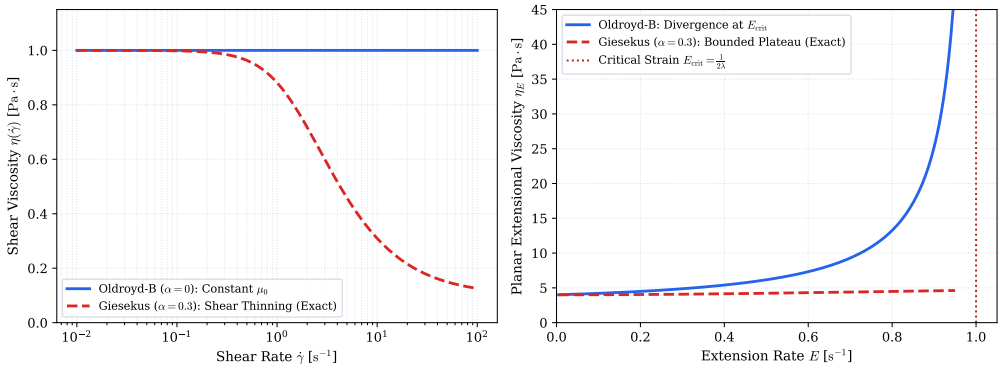


Figure 2.1: Example comparison of rheological model responses: (a) steady shear viscosity $\eta(\dot{\gamma})$ as a function of shear rate, illustrating the constant Newtonian-like plateau of Oldroyd-B ($\alpha = 0$) versus the non-linear shear-thinning response of Giesekus ($\alpha > 0$); (b) planar extensional viscosity $\eta_E(E)$ as a function of extension rate, contrasting the unphysical asymptotic divergence of Oldroyd-B at the critical strain threshold $E_{\text{crit}} = 1/(2\lambda)$ with the bounded saturation plateau provided by non-linear closures.

2.4.1 Reference Scales and Dimensionless Parameters

We choose reference length H_{ref} (the roller radius R), reference velocity U_{ref} (the roller surface peripheral speed u_{roll}), and a characteristic dynamic viscosity scale η_0 (taken as the baseline total viscosity μ_0). The dimensionless variables are defined as:

$$\boldsymbol{x}^* = \frac{\boldsymbol{x}}{H_{\text{ref}}}, \quad \boldsymbol{u}^* = \frac{\boldsymbol{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \tag{2.25}$$

Dropping asterisks for notational simplicity, the flow regime is governed by three fundamental dimensionless groups:

$$Re = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \qquad \text{(Reynolds number)}, \tag{2.26}$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \qquad \text{(Weissenberg number)}, \tag{2.27}$$

$$\beta = \frac{\mu_s}{\mu_0} = \frac{\mu_s}{\mu_s + \mu_p} \qquad \text{(Viscosity ratio)}. \tag{2.28}$$

Each dimensionless group carries a precise physical meaning:

- **Reynolds number (Re)**: Compares convective inertial forces to viscous dissipation forces. In the present benchmark, $Re \ll 1$, placing the system firmly in the laminar creeping flow (Stokes) regime.

- **Weissenberg number (Wi)**: Compares the characteristic elastic relaxation time of the polymer λ to the characteristic kinematic timescale of the flow $H_{\text{ref}}/U_{\text{ref}}$. It quantifies the degree of flow-induced macromolecular orientation and elastic non-linearity.

- **Viscosity ratio (β)**: Measures the fraction of total viscosity contributed by the Newtonian solvent. The limits $\beta = 1$ and $\beta \to 0$ correspond to a purely Newtonian fluid and a solvent-free polymer melt, respectively. The dimensionless solvent and polymeric viscosities are given by $\mu_s^* = \beta$ and $\mu_p^* = 1 - \beta$.

2.4.2 Dimensionless Governing System

Expressed in terms of dimensionless variables, the steady planar governing equations become:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2.29}$$

$$Re(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) + \nabla p - \beta \nabla^2 \boldsymbol{u} - \nabla \cdot \boldsymbol{\tau} = \boldsymbol{0}, \tag{2.30}$$

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) + Wi\left(\boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T\right) - 2(1-\beta)\boldsymbol{D} = \boldsymbol{0}, \tag{2.31}$$

where the constitutive operator $\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ provides a unified closure for the models under investigation:

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B)}, \\ \boldsymbol{\tau} + \frac{\alpha Wi}{1-\beta}\boldsymbol{\tau}^2 & \text{(Giesekus)}, \\ \left(1 + \frac{\varepsilon Wi}{1-\beta}\mathrm{tr}(\boldsymbol{\tau})\right)\boldsymbol{\tau} & \text{(Linear PTT)}. \end{cases} \tag{2.32}$$

2.5 Four-Roll Mill

The four-roll mill is a classic benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally introduced by Taylor [5] and systematically analyzed by Fuller and Leal [7, 17], it provides an ideal apparatus for investigating macromolecular deformation away from solid walls. As illustrated conceptually in Figure 2.2, the setup comprises four cylindrical rollers placed symmetrically inside a closed square cavity.

2.5.1 Flow Index Parameter

In the central region bounded by the four rollers, the velocity field can be represented as a linear superposition of extension and solid-body rotation [7, 17]:

$$\boldsymbol{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \tag{2.33}$$

where $E \geq 0$ is the extension rate and $\omega \geq 0$ is the angular rotation rate. To quantify the relative magnitude of extension versus rotation, we define a dimen-

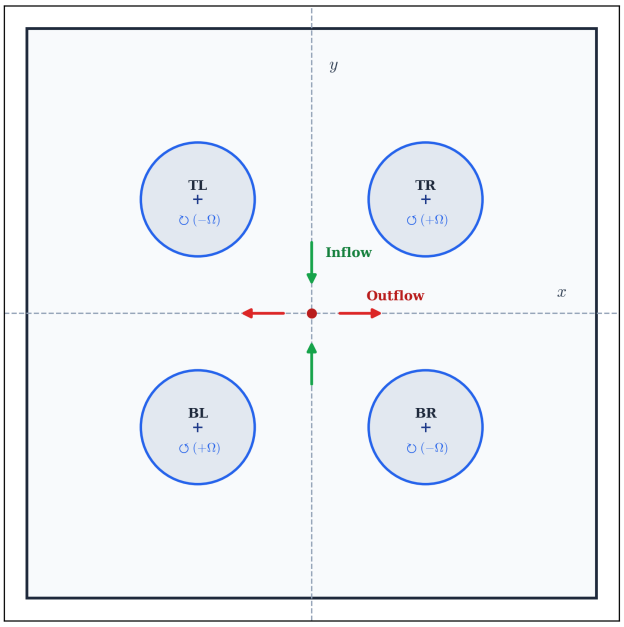


Figure 2.2: Conceptual diagram of the four-roll mill geometry, illustrating the four rotating cylinders (adjacent rolls counter-rotate with angular velocity Ω, diagonal rolls co-rotate), the central hyperbolic stagnation point $(0, 0)$, and the resulting elongational inflow and outflow directions.

sionless flow type parameter $\chi \in [-1, 1]$:

$$\chi = \frac{E - \omega}{E + \omega}. \tag{2.34}$$

Depending on roller rotation speeds and directions, this local kinematics covers three canonical flow regimes (in Figure 2.3):

- $\chi = +1$ $(\omega = 0)$: **Pure Planar Extensional Flow**. Generated when adjacent rollers counter-rotate symmetrically (diagonal rollers co-rotate). Fluid elements experience pure elongation along x and compression along y, producing hyperbolic streamlines $xy = \text{const}$.

- $\chi = 0$ $(E = \omega)$: **Simple Shear Flow**. Characterized by equal strain and rotation rates, producing rectilinear shear streamlines locally near the origin.

- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation**. Generated when rollers rotate in unison to drive bulk cavity circulation, resulting in concentric circular streamlines without stretching.

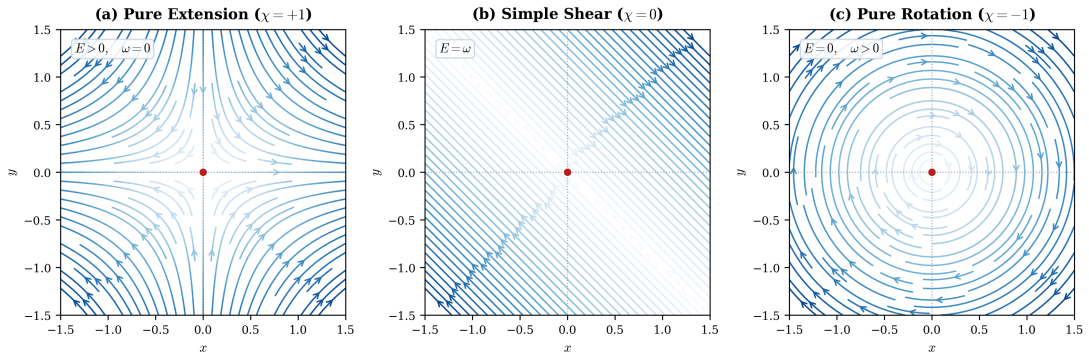


Figure 2.3: Streamline patterns of the two-dimensional linear flow field in Equation (2.33) for the three canonical regimes of the flow classification parameter χ: (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

2.5.2 Stagnation Point

In the planar extensional regime ($\chi = +1$), the velocity vanishes identically at the cavity center:

$$\boldsymbol{u}(0, 0) = \mathbf{0}. \tag{2.35}$$

Along the vertical inflow streamline ($x = 0$), incoming fluid parcels decelerate toward zero velocity, reaching asymptotically long residence times near the origin. This kinematics allows elastic strain to accumulate over durations comparable to or exceeding the polymer relaxation time λ, making the four-roll mill an ideal benchmark for probing strong extensional stresses far from solid walls.

2.5.3 Geometric Setup

Following the established benchmark setup of D'Avino et al. [18], we define the physical dimensions of the apparatus:

- **Square cavity**: Side length $L = 50\,\text{mm}$ spanning the domain $[-25, 25]\,\text{mm}$.
- **Cylindrical rollers**: Radius $R = 5\,\text{mm}$ ($H_{\text{ref}} = R = 0.1L$).
- **Roller centers**: Located symmetrically at $(\pm 10, \pm 10)\,\text{mm}$ ($d_{\text{roll}} = 10\,\text{mm}$).
- **Dimensionless domain**: In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

As shown in Figure 2.4, these dimensions define the operational space and solid boundaries of the cavity.

2.5.4 Operating Parameters

The flow is driven by the rotation of the four cylindrical rollers with peripheral surface velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}\,\text{m/s}$. The geometric, fluid physical, and characteristic dimensionless parameters investigated across the benchmark test cases are organized systematically in Table 2.1.

2.5.5 Boundary Conditions

Let Ω denote the planar fluid domain bounded externally by the square cavity walls Γ_{outer} and internally by the boundaries of the four cylindrical rollers $\Gamma_{\text{rolls}} = \bigcup_k \Gamma_k$ ($k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$), such that the total boundary is given by the union $\partial\Omega = \Gamma_{\text{outer}} \cup \Gamma_{\text{rolls}}$, as illustrated in Figure 2.4.

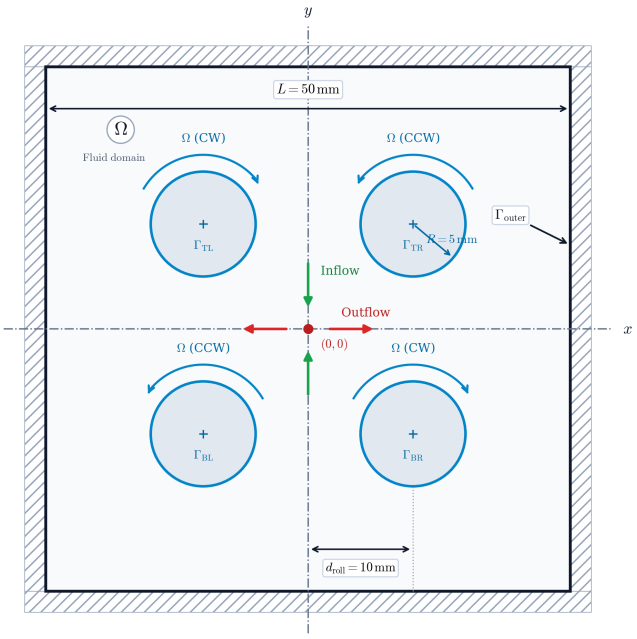


Figure 2.4: Detailed geometric dimensions and boundary partition of the four-roll mill: planar fluid domain Ω, external stationary cavity boundary Γ_{outer} with side length $L = 50\,\mathrm{mm}$, four cylinder boundaries $\Gamma_{\mathrm{rolls}} = \bigcup_k \Gamma_k$ ($k \in \{\mathrm{TL}, \mathrm{TR}, \mathrm{BL}, \mathrm{BR}\}$) with radius $R = 5\,\mathrm{mm}$ and center offset $d_{\mathrm{roll}} = 10\,\mathrm{mm}$, peripheral rotation directions with angular velocity Ω, and central extensional stagnation point $(0, 0)$.

Table 2.1: Geometric, physical, and dimensionless parameters of the four-roll mill benchmark.

Parameter	**Symbol**	**Value / Range**	**Unit**
Geometric and Kinematic Setup			
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\mathrm{ref}} = R$	0.005	m
Roller center offset	d_{roll}	0.010	m
Roller peripheral velocity	$u_{\mathrm{roll}} = U_{\mathrm{ref}}$	8.333×10^{-3}	m/s
Fluid Physical Properties			
Fluid density	ρ	1000	$\mathrm{kg/m^3}$
Characteristic viscosity scale	$\eta_0 = \mu_0$	1.0	$\mathrm{Pa \cdot s}$
Solvent dynamic viscosity	μ_s	0.1–0.5	$\mathrm{Pa \cdot s}$
Polymer dynamic viscosity	μ_p	0.5–0.9	$\mathrm{Pa \cdot s}$
Polymer relaxation time	λ	0.05–1.0	s
Dimensionless Governing Groups			
Reynolds number	Re	$0.0417 \ll 1$	–
Weissenberg number	Wi	0.083–1.667	–
Viscosity ratio	β	0.1–0.5	–

To close the dimensionless governing system (2.29)–(2.31), kinematic Dirichlet conditions and a pressure reference point are prescribed:

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{0}, \qquad \boldsymbol{x} \in \Gamma_{\text{outer}}, \tag{2.36}$$

$$\boldsymbol{u}(\boldsymbol{x}) = \boldsymbol{\Omega}_k \times (\boldsymbol{x} - \boldsymbol{x}_k), \qquad \boldsymbol{x} \in \Gamma_k \quad (k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}), \tag{2.37}$$

$$p(\boldsymbol{x}_0) = 0, \qquad \boldsymbol{x}_0 \in \Omega. \tag{2.38}$$

In the planar extensional regime ($\chi = +1$), the roller angular velocity vectors $\boldsymbol{\Omega}_k = (0, 0, \Omega_k)^T$ are oriented along the out-of-plane z-axis such that diagonal pairs co-rotate and adjacent pairs counter-rotate:

$$\Omega_{\text{TL}} = -\Omega, \quad \Omega_{\text{TR}} = +\Omega, \quad \Omega_{\text{BL}} = +\Omega, \quad \Omega_{\text{BR}} = -\Omega, \tag{2.39}$$

with peripheral angular speed $\Omega = u_{\text{roll}}/R$.

Because fluid cannot penetrate the impermeable solid boundaries ($\boldsymbol{u} \cdot \boldsymbol{n} = 0$ on $\partial\Omega$), the global mass conservation constraint $\oint_{\partial\Omega} \boldsymbol{u} \cdot \boldsymbol{n} \, \mathrm{d}\Gamma = 0$ is satisfied identically. The condition (2.38) anchors the hydrostatic pressure field. Finally, because the four-roll mill constitutes a completely closed recirculating cavity without inflow boundaries ($\Gamma_{\text{in}} = \emptyset$), no external Dirichlet conditions are mathematically required for the extra-stress tensor $\boldsymbol{\tau}$.

Chapter 3

Physics-Informed Neural Networks in Complex Rheology

Feedforward deep neural networks, commonly known as Multi-Layer Perceptrons (MLPs), provide flexible continuous approximations for non-linear functions [15]. An MLP is formed by composing a sequence of parametric functions, called *layers*. Each layer maps its inputs through a linear transformation followed by a non-linear operation. With these layers in series, the network can represent continuous functional mappings of arbitrary complexity without requiring explicit basis functions.

In scientific computing, neural networks represent physical fields directly as continuous functions over spatial coordinates, rather than storing discrete state variables on a mesh. However, using purely data-driven networks for fluid mechanics presents clear limits: they require large amounts of labeled data, can violate conservation laws, and cannot infer latent material properties on their own.

Physics-Informed Neural Networks (PINNs) address these limits by embedding governing partial differential equations directly into the training loss. The differential system formulated in Chapter 2, governed by mass conservation, vis-

coelastic momentum balance, and frame-indifferent constitutive transport, represents a coupled set of non-linear equations that poses well-known challenges to traditional mesh-based methods. In this chapter, we formulate this continuous boundary-value problem within a Scientific Machine Learning (SciML) framework. By embedding momentum and constitutive balances into the optimization objective through automatic differentiation, the PINN framework enables both forward state simulation and inverse rheological parameter discovery without spatial discretization meshes.

3.1 Fundamentals of Deep Learning

Deep learning models approximate continuous mathematical functions by learning parameters directly from data [15]. In this section, we review the fundamental building blocks of feedforward networks: the artificial neuron, activation functions, multi-layer architectures, loss functions, optimization algorithms, and gradient computation.

3.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron. Given an input vector $\boldsymbol{x}$, a neuron computes a linear combination of the inputs followed by a non-linear activation:

$$z = \boldsymbol{w} \cdot \boldsymbol{x} + b, \qquad a = \sigma(z), \tag{3.1}$$

where $\boldsymbol{w}$ is the weight vector, b is a scalar bias, $\sigma(\cdot)$ is the activation function and a is the output. Non-linear activations are essential: without them, composing multiple linear layers simply collapses into a single linear transformation.

In classical machine learning, piecewise-linear functions such as the Rectified Linear Unit, $\mathrm{ReLU}(z) = \max(0, z)$, are widely favored for their computational efficiency. However, in scientific computing and fluid mechanics, the network outputs parameterize continuous physical fields governed by differential equations. These governing balance laws involve spatial derivatives of first, second, or third order, such as viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ or stress divergence $\nabla \cdot \boldsymbol{\tau}$.

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere, evaluating a ReLU network inside differential residuals eliminates diffusive and curvature terms. Physics-informed models therefore require smooth, differentiable activation functions ($\mathcal{C}^2$ or higher). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [15]:

$$\mathrm{SiLU}(z) = \frac{z}{1 + e^{-z}}. \tag{3.2}$$

The SiLU activation is smooth and non-monotonic. Unlike the hyperbolic tangent (tanh), which saturates toward constant values for large positive inputs, SiLU remains active in the positive regime. This property maintains healthy gradient propagation in flow regions with intense shear and steep localized stress gradients, which are characteristic of viscoelastic flows.

3.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector $\boldsymbol{x}$ to an output prediction

$\hat{\boldsymbol{u}}$ by propagating information through L successive layers [15]:

$$\begin{aligned} \boldsymbol{h}^{(1)} &= \sigma\left(\boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)}\right), \\ \boldsymbol{h}^{(l)} &= \sigma\left(\boldsymbol{W}^{(l)}\boldsymbol{h}^{(l-1)} + \boldsymbol{b}^{(l)}\right), \quad \text{for } l = 2, \ldots, L-1, \\ \hat{\boldsymbol{u}} &= \boldsymbol{W}^{(L)}\boldsymbol{h}^{(L-1)} + \boldsymbol{b}^{(L)}, \end{aligned} \tag{3.3}$$

where $\boldsymbol{h}^{(l)}$ denotes the hidden feature vector at layer l, and $\boldsymbol{W}^{(l)}$ and $\boldsymbol{b}^{(l)}$ represent the weight matrix and bias vector of that layer. In our implementation, the inputs $\boldsymbol{x} = (x, y)$ are spatial coordinates within the fluid cavity, while the outputs $\hat{\boldsymbol{u}}$ parameterize the physical state variables: the stream function ψ, hydrodynamic pressure p, and extra-stress tensor components $\boldsymbol{\tau}$.

A central theoretical property of MLPs is universal function approximation [14,23]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a compact domain to arbitrary precision. However, universal approximation guarantees only the existence of a suitable set of weights, not that standard gradient-based optimization will automatically find them or that spatial derivatives will match the physical solution with equal fidelity.

Importance of Weight Initialization

The initialization of network parameters strongly conditions the resulting optimization dynamics. Because deep architectures compose multiple transformations and non-linearities, inappropriate initial weight magnitudes can cause layer activations and backpropagated gradients to either vanish or explode across depth [15]. In scientific computing and differential problems, where non-linear convective terms and tensor couplings are evaluated, uncalibrated initial weights can trigger extreme initial residual spikes that immediately destabilize gradient

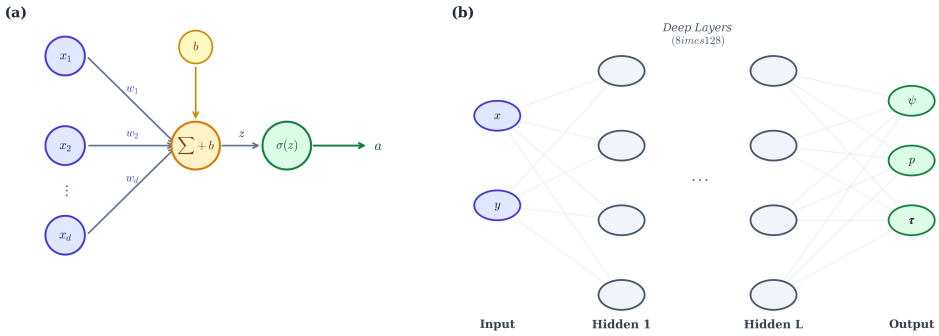


Figure 3.1: Neural network representations: (a) schematic of an artificial neuron computing an affine combination $z = \boldsymbol{w} \cdot \boldsymbol{x} + b$ followed by a smooth activation $\sigma(z)$; (b) deep Multi-Layer Perceptron (MLP) architecture propagating coordinate inputs (x, y) through L hidden layers to parameterize continuous physical fields $(\psi, p, \boldsymbol{\tau})$.

descent. Appropriate variance-preserving initialization schemes, such as scaling parameter distributions inversely with layer widths, help preserve stable gradient flow and avoid optimization divergence. Specific configuration choices regarding hidden and output layer initialization adopted in this work are detailed in Chapter 4.

3.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of identifying a parameter set $\boldsymbol{\theta} = \{\boldsymbol{W}^{(l)}, \boldsymbol{b}^{(l)}\}$ that minimizes a scalar objective function $\mathcal{L}(\boldsymbol{\theta})$. In supervised regression, the discrepancy between model predictions and target reference measurements is commonly quantified through the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i; \boldsymbol{\theta}) - \boldsymbol{u}_i^{\text{target}} \right\|_2^2 . \tag{3.4}$$

Parameters are iteratively updated along descent directions derived from the loss gradient:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{d}_k, \tag{3.5}$$

where $\eta > 0$ represents the learning rate (step size) and $\boldsymbol{d}_k$ denotes the search direction at iteration k.

In scientific machine learning, gradient-based solvers must navigate non-convex loss surfaces characterized by ill-conditioned curvature and local minima. Two widely adopted algorithms in this setting are:

First-Order Adaptive Moment Estimation (Adam)

The Adam optimizer [22] computes individual adaptive learning rates for each network parameter by tracking running estimates of the first and second moments of the loss gradient. At iteration k, given the gradient $\boldsymbol{g}_k = \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}_k)$, the algorithm updates two vectors recursively:

$$\boldsymbol{m}_k = \beta_1 \boldsymbol{m}_{k-1} + (1 - \beta_1)\boldsymbol{g}_k, \tag{3.6}$$

$$\boldsymbol{v}_k = \beta_2 \boldsymbol{v}_{k-1} + (1 - \beta_2)\boldsymbol{g}_k^2, \tag{3.7}$$

where the square $\boldsymbol{g}_k^2$ is evaluated component-wise, and $\beta_1, \beta_2 \in [0, 1)$ are exponential decay factors (canonically set to $\beta_1 = 0.9$ and $\beta_2 = 0.999$).

These two vectors have distinct physical and optimization roles:

- **First moment vector $\boldsymbol{m}_k$ (Directional Momentum)**: maintains an exponentially decaying average of past gradients. It acts as a directional inertia, accelerating descent along persistent slopes while dampening high-frequency oscillations across walls.

- **Second uncentered moment vector $\boldsymbol{v}_k$ (Adaptive Scaling)**: tracks the running magnitude of squared gradient components. By dividing the update step by $\sqrt{\boldsymbol{v}_k}$, Adam rescales the effective learning rate coordinate by coordinate, taking cautious smaller steps along steep, noisy directions and bolder steps along flat plateaus.

Because both moment vectors are initialized to zero ($\boldsymbol{m}_0 = \boldsymbol{v}_0 = \boldsymbol{0}$), they initially exhibit a numerical bias toward zero. Kingma and Ba [22] rescale $\boldsymbol{m}_k$ and $\boldsymbol{v}_k$ by $(1-\beta_1^k)$ and $(1-\beta_2^k)$ to correct this initial bias before updating the parameter vector $\boldsymbol{\theta}_{k+1}$. This adaptive mechanism makes Adam particularly robust on rugged, multi-objective loss landscapes characterized by disparate gradient scales, serving as an effective global explorer in early training.

Second-Order Quasi-Newton Optimization (L-BFGS)

While first-order methods explore the loss landscape robustly, refining physical solutions and satisfying stiff differential equations requires incorporating second-order curvature information. Classical Newton methods require forming and inverting the full Hessian matrix $\nabla^2_{\boldsymbol{\theta}}\mathcal{L}$, which is computationally prohibitive for neural networks containing $N \sim 10^5$–10^6 parameters ($\mathcal{O}(N^2)$ memory and $\mathcal{O}(N^3)$ inversion cost).

Quasi-Newton methods overcome this limitation by iteratively approximating the inverse Hessian matrix $\boldsymbol{H}_k \approx (\nabla^2_{\boldsymbol{\theta}}\mathcal{L})^{-1}$ using only successive parameter displacements $\boldsymbol{s}_k = \boldsymbol{\theta}_{k+1} - \boldsymbol{\theta}_k$ and gradient variations $\boldsymbol{y}_k = \nabla_{\boldsymbol{\theta}}\mathcal{L}_{k+1} - \nabla_{\boldsymbol{\theta}}\mathcal{L}_k$. The classic BFGS update formula is given by [25]:

$$\boldsymbol{H}_{k+1} = (\boldsymbol{I} - \rho_k \boldsymbol{s}_k \boldsymbol{y}_k^\top)\boldsymbol{H}_k(\boldsymbol{I} - \rho_k \boldsymbol{y}_k \boldsymbol{s}_k^\top) + \rho_k \boldsymbol{s}_k \boldsymbol{s}_k^\top, \qquad \rho_k = \frac{1}{\boldsymbol{y}_k^\top \boldsymbol{s}_k}. \tag{3.8}$$

To avoid storing the dense $N \times N$ matrix $\boldsymbol{H}_{k+1}$, the Limited-memory BFGS (L-BFGS) algorithm [25,26] stores only the m most recent vector pairs $\{\boldsymbol{s}_i, \boldsymbol{y}_i\}_{i=k-m}^{k-1}$ (typically $m \in [10, 50]$). The descent direction is then evaluated on the fly via an efficient two-loop recursion algorithm, dramatically reducing the memory footprint from $\mathcal{O}(N^2)$ to $\mathcal{O}(mN)$. The step length along this direction is calibrated through a line search ensuring sufficient decrease of the objective function. When evaluated over full collocation batches in double precision (FP64), L-BFGS ex-

hibits fast superlinear asymptotic convergence, making it ideal for the final refinement stage of PINN solvers.

3.1.4 Backpropagation

To update network parameters via gradient descent, we compute the partial derivatives of the scalar loss with respect to all weights and biases, $\nabla_{\boldsymbol{\theta}}\mathcal{L}$. Evaluating these gradients through individual numerical perturbations would be computationally prohibitive.

Instead, neural networks compute analytical gradients using the *Automatic Differentiation (AD)* algorithm [16]. Automatic differentiation evaluates derivatives to machine precision by applying the chain rule to elementary operations along the computational graph. In standard machine learning, reverse-mode AD is widely known as the *backpropagation* algorithm, primarily employed to evaluate the gradient of the scalar loss with respect to network parameters.

This reverse-mode automatic differentiation evaluates all parameter gradients simultaneously at a computational cost comparable to a single forward pass, making deep network training feasible.

3.2 Neural Networks: Capabilities and Physical Limitations

3.2.1 Key Advantages

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **High-Dimensional Functional Approximation**: The universal approximation capacity allows neural networks to represent complex, coupled phys-

ical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.

2. **Meshless Representation**: Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a continuous, globally defined mapping $\boldsymbol{x} \mapsto \hat{\boldsymbol{u}}(\boldsymbol{x})$ over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being meshless does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.

3. **Hardware Acceleration and Differentiability**: Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

3.2.2 Limits of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven black-box networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

- **Data scarcity in experimental rheology**: Deep neural networks typically rely on large labeled datasets. In fluid experiments, acquiring dense velocity fields via Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are very hard to obtain in complex geometries.

- **Absence of physical grounding**: Purely data-driven models minimize empirical loss on discrete observations without physical constraints. Consequently, their predictions often violate incompressibility ($\nabla \cdot \boldsymbol{u} \neq 0$), momentum balance, and constitutive laws, generating non-physical artifacts when interpolating or extrapolating.

- **Overfitting**: In sparse-data regimes, over-parameterized networks can easily fit measurement noise, obtaining low training errors while losing accuracy in unmonitored regions.

- **Inability to discover governing parameters**: Black-box models learn input-output mappings, but cannot extract underlying physical properties, such as relaxation times or polymeric viscosities, that govern flow dynamics.

These limitations show that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed physical conservation laws directly into the loss function.

3.3 Physics-Informed Neural Network

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [8], addresses the problems of purely data-driven modeling by incorporating differential partial equations directly into the network training objective.

3.3.1 Physics-Constrained Loss Formulation

Following the unified boundary-value problem (BVP) notation established in the scientific machine learning literature [24], consider a physical state described by a vector of continuous fields $\boldsymbol{u}(\boldsymbol{x})$ defined over a spatial domain $\Omega \subset \mathbb{R}^d$ bounded

by $\partial\Omega$:

$$\mathcal{F}[\boldsymbol{u}(\boldsymbol{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \boldsymbol{x} \in \Omega, \tag{3.9}$$

$$\mathcal{B}[\boldsymbol{u}(\boldsymbol{x})] = \boldsymbol{g}(\boldsymbol{x}), \quad \boldsymbol{x} \in \partial\Omega, \tag{3.10}$$

where:

- $\boldsymbol{u}(\boldsymbol{x})$ denotes the physical state variables (in our viscoelastic framework, $\boldsymbol{u} = (\psi, p, \boldsymbol{\tau})$, comprising stream function, pressure, and extra-stress tensor components);
- $\mathcal{F}$ is a general non-linear differential operator encoding the governing field equations (mass conservation, momentum balance, and constitutive transport equations);
- $\boldsymbol{\lambda}$ denotes physical or constitutive parameters governing the flow dynamics (such as solvent viscosity, polymeric viscosity, and relaxation time);
- $\mathcal{B}$ denotes the boundary operator enforcing kinematic constraints on the boundaries $\partial\Omega$ (such as no-slip conditions along solid walls and prescribed angular velocities on roller surfaces), with prescribed boundary values $\boldsymbol{g}(\boldsymbol{x})$.

In a PINN, the continuous solution fields are parameterized by a deep neural network surrogate $\hat{\boldsymbol{u}}(\boldsymbol{x}; \boldsymbol{\theta})$ parameterized by weights and biases $\boldsymbol{\theta}$. Evaluating the governing differential operator on the network outputs via automatic differentiation defines the pointwise differential residual:

$$\boldsymbol{f}(\boldsymbol{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\boldsymbol{u}}(\boldsymbol{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \tag{3.11}$$

The network parameters $\boldsymbol{\theta}$, alongside any unknown physical parameters $\boldsymbol{\lambda}$ in inverse problems, are optimized simultaneously by minimizing a composite multi-

objective loss:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \tag{3.12}$$

where each individual loss component is evaluated as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i^{\text{data}}) - \boldsymbol{u}_i^{\text{data}} \right\|_2^2, \tag{3.13}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\boldsymbol{u}}(\boldsymbol{x}_j^{\text{bc}})] - \boldsymbol{g}(\boldsymbol{x}_j^{\text{bc}}) \right\|_2^2, \tag{3.14}$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \boldsymbol{f}(\boldsymbol{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \tag{3.15}$$

Here, $\{\boldsymbol{x}_i^{\text{data}}\}$ denotes available empirical observations or measurement locations, $\{\boldsymbol{x}_j^{\text{bc}}\}$ represents points sampled along physical boundaries $\partial\Omega$, and $\{\boldsymbol{x}_k^{\text{pde}}\}$ designates interior *collocation points* scattered across Ω.

This formulation naturally supports two complementary problem settings:

- **Forward problems**: The physical parameters $\boldsymbol{\lambda}$ are known. The network optimizes its weights $\boldsymbol{\theta}$ to solve for the continuous fields $\hat{\boldsymbol{u}}(\boldsymbol{x})$, acting as a meshless continuous PDE solver.

- **Inverse problems**: One or more constitutive parameters in $\boldsymbol{\lambda}$ are unknown. The network simultaneously reconstructs the unobserved latent fields and identifies the unknown parameters by combining empirical velocity observations $\mathcal{L}_{\text{data}}$ with physical consistency constraints $\mathcal{L}_{\text{pde}}$.

A key methodological principle of this thesis is the distinction between field reconstruction accuracy and parameter identifiability. A network may fit observed velocity fields accurately, yet inferred rheological parameters may remain uncertain if the kinematics lack sufficient variation or if the loss landscape exhibits degeneracies. Embedding physical equations provides essential structure,

but does not automatically guarantee unique parameter recovery.

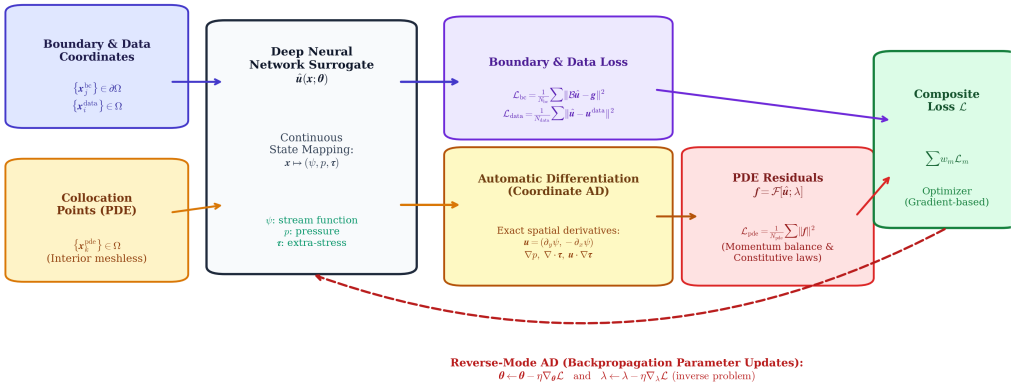


Figure 3.2: Schematic computational workflow of a Physics-Informed Neural Network (PINN): boundary coordinates $\{\boldsymbol{x}_j^{\text{bc}}\}$ and measurements $\{\boldsymbol{x}_i^{\text{data}}\}$ evaluate boundary adherence ($\mathcal{L}_{\text{bc}}$) and empirical matching ($\mathcal{L}_{\text{data}}$) (top branch), while interior collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$ pass through Coordinate Automatic Differentiation (AD) to compute exact spatial derivatives and differential PDE residuals ($\mathcal{L}_{\text{pde}}$, bottom branch). The composite loss $\mathcal{L}_{\text{total}}$ drives gradient descent via reverse-mode backpropagation, jointly optimizing surrogate weights $\boldsymbol{\theta}$ and identifying unknown physical parameters $\boldsymbol{\lambda}$ in inverse settings.

3.3.2 PINN-Specific Features

Several operational features distinguish physics-informed training from conventional supervised learning:

1. **Training as a boundary-value solver**: In classical machine learning, a model is trained on a sample set to generalize to unseen statistical distributions. In PINNs, *training is the numerical solver*. The network is not designed to predict flows across arbitrary unseen geometries; rather, optimization directly solves the specific problem on the specified domain Ω. Available measurements $\mathcal{L}_{\text{data}}$ act as an active data anchor that guides the network toward the true physical state.

2. **Physical regularization and dynamic loss weighting**: In standard

supervised training, the network is unconstrained between data points. In PINNs, evaluating differential residuals $\mathcal{L}_{\text{pde}}$ across interior collocation points penalizes non-physical gradients and curvatures, restricting the admissible solution space. However, balancing the competing loss weights (w_{data}, w_{bc}, and w_{pde}) presents practical challenges. As analyzed by Wang, Teng, and Perdikaris [12], the backpropagated gradient magnitudes arising from PDE residuals and boundary constraints often differ by several orders of magnitude. This imbalance induces numerical stiffness, where boundary terms can dominate and suppress the learning signal of interior PDE residuals (or vice versa). Calibrated weighting between data, boundary, and differential components is therefore needed to prevent optimization stalls and ensure stable convergence.

3. **Coordinate Automatic Differentiation (AD)**: Automatic differentiation serves a dual purpose in PINNs: beyond computing loss gradients with respect to network parameters, it evaluates exact spatial derivatives of the predicted physical fields with respect to coordinates:

$$\text{Coordinate AD (PINNs):} \quad \frac{\partial \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}}, \quad \frac{\partial^2 \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}^2}. \tag{3.16}$$

The accuracy of these derivatives with respect to the true physical field depends on how closely the network itself approximates the physical solution.

4. **Collocation point sampling**: Unlike traditional CFD solvers that require meshes and element connectivity tables, PINNs evaluate differential residuals at sparse collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$. This meshless feature simplifies domain representation, though sufficient point density remains necessary in boundary layers and near stagnation regions.

Table 3.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	Discrete difference quotients (Taylor series)	$\mathcal{O}(\Delta x^2)$	Structured Mesh Stencil
Symbolic Differentiation	Closed-form analytical expression	None	Expression Growth
Automatic Differentiation (AD)	Exact chain rule on computational graph	**None (Machine Precision)**	**Meshless / Continuous**

3.4 Foundational Literature Benchmarks

3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational paper, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capability of PINNs on laminar vortex past a cylinder at $Re = 100$.

Problem Formulation

The dimensionless incompressible Navier-Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \tag{3.17}$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \tag{3.18}$$

$$u_x + v_y = 0, \tag{3.19}$$

where $\lambda_1 = 1.0$ governs convective acceleration and $\lambda_2 = 1/Re = 0.01$ represents kinematic viscosity.

To satisfy incompressibility identically without requiring an explicit divergence penalty, the authors adopted the stream function formulation established in Section 2.2.1 (Eq. 2.10), parameterizing velocity components through spatial derivatives of $\psi(t, x, y)$. A neural network with 9 hidden layers and 20 neurons per layer, activated by tanh, predicted (ψ, p) while treating λ_1 and λ_2 as learnable parameters.

Field and Parameter Identification

The network was trained on $N = 5{,}000$ spatio-temporal velocity points. The benchmark demonstrated two key capabilities:

- **Pressure recovery**: Without any pressure measurements, the network reconstructed the pressure field $p(t, x, y)$ exclusively relying on momentum balance, matching CFD reference data up to an arbitrary constant.
- **Parameter estimation under noise**: On clean data, parameter errors were 0.08% for λ_1 and 4.67% for λ_2. Under 1% uncorrelated Gaussian noise, errors remained controlled at 0.17% and 5.70%.

While this benchmark validated simultaneous field reconstruction and parameter identification, it was restricted to linear Newtonian fluids in an open domain.

3.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending physics-informed methods to non-Newtonian fluids, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, applying PINNs to polymeric flows governed by differential constitutive equations.

The Rheological Bottleneck

Viscoelastic fluid simulation presents two major challenges:

1. **Experimental limitations**: Non-intrusive measurements of internal extra-stresses $\boldsymbol{\tau}$ are generally unavailable in complex geometries.
2. **Numerical instability (HWNP)**: In standard grid-based solvers (FEM, FVM), elevated elasticity triggers the High-Weissenberg Number Problem (HWNP), where steep stress boundary layers cause convergence breakdown unless stabilized numerically.

Virtual Rheometer

To address these limits, ViscoelasticNet proposed an inverse framework operating as a "virtual rheometer":

- **Stress inference**: Given velocity measurements $\boldsymbol{u}$, the network infers unobserved extra-stresses $\boldsymbol{\tau}$ and pressure p by enforcing constitutive and momentum residuals.
- **Unified constitutive formulation**: The authors formulated a unified constitutive expression encompassing Oldroyd-B, linear PTT, and Giesekus models:
$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}(\boldsymbol{\tau}\cdot\boldsymbol{\tau}) + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau})\boldsymbol{\tau} = 2\mu_p \boldsymbol{D}, \tag{3.20}$$
seeking to identify λ, μ_p, α, and ε.
- **Decoupled two-stage training**: Optimization was separated into two sequential stages:
  - *Stage 1 (Stress and Rheology)*: Velocity and stress networks were optimized jointly using velocity data and constitutive residuals, learning rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
  - *Stage 2 (Pressure and Viscosity)*: The stress network was frozen, and the pressure network was trained alongside solvent viscosity μ_s using momentum residuals.
- **Investigated geometries**: ViscoelasticNet was evaluated on two open geometries: a planar stenosed channel and a cross-slot flow. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These results confirmed that PINNs can infer non-Newtonian stresses from velocity fields in open channel flows. However, their formulation focused on do-

mains with net through-flow and convective pressure drops. In the next chapter, we address the open problem of enclosed recirculating flows: we formulate a stream-function PINN for the four-roll mill geometry, evaluate forward field accuracy, and investigate the boundaries of inverse parameter discovery.

Chapter 4

Methodology and Numerical Results

The previous chapters established the governing equations of viscoelastic flow (Chapter 2) and the physics-informed neural network framework (Chapter 3). In this chapter, we evaluate that framework on the four-roll mill benchmark. The goal is twofold: first, to test the forward predictive solver against high-fidelity numerical solutions; second, to assess the inverse PINN as a virtual rheometer capable of reconstructing hidden stress and pressure fields while extracting unknown constitutive parameters directly from kinematic observations.

We first present the problem formulation in the recirculating cavity, the reference finite element dataset generated with COMSOL Multiphysics, and the decoupled network architecture. We then evaluate forward field predictions, report inverse discovery results across Oldroyd-B, Giesekus, and PTT fluids, and examine the physical limits, invariances, and degeneracies that govern viscoelastic parameter estimation.

4.1 Problem Formulation

4.1.1 Recirculating Cavity Challenges

Most benchmark studies in scientific machine learning for non-Newtonian flows focus on open channels with net through-flow, such as planar contractions or cross-slots driven by pressure gradients. The four-roll mill [18] is fundamentally different: it is a closed square cavity driven solely by four rotating cylinders, generating recirculating streamlines with zero net mass flux.

This setup presents specific computational and physical challenges:

- **Coexistence of Extensional and Shearing Regimes**: The flow couples a planar extensional hyperbolic stagnation point at the domain center with intense simple shear confined within the narrow gaps between the rotating rollers and the outer walls.

- **Stress Recirculation along Closed Streamlines**: Polymeric extra-stresses advect along closed trajectories. Because there are no open inlets or outlets to carry away convective transients, the neural network must resolve localized stress boundary layers and thin strands without introducing spurious stress accumulation or artificial numerical diffusion.

- **High-Weissenberg Regimes as a Stress Test**: While many previous inverse studies restrict attention to small relaxation times ($\lambda \leq 0.15\,\mathrm{s}$), we intentionally test higher values (λ up to $1.0\,\mathrm{s}$, corresponding to $Wi > 1$). This pushes the solver into regimes where convective stress derivatives become stiff and nonlinear elastic effects challenge numerical stability.

4.1.2 Forward and Inverse Modes

The computational framework operates in two distinct modes:

- **Direct Mode (Latent Field Reconstruction)**: All constitutive fluid parameters (λ, μ_p, μ_s, and nonlinear coefficients α or ε) are assumed known from prior characterization. Planar velocity observations (representative of experimental PIV data) supervise the stream function $\hat{\psi}$ in the interior alongside solid boundary conditions. The network then autonomously reconstructs the unmeasured latent extra-stress tensor $\hat{\boldsymbol{\tau}}(x, y)$ and hydrodynamic pressure $\hat{p}(x, y)$ enforcing the momentum and constitutive residuals without any interior stress or pressure supervision.

- **Inverse Mode (Virtual Rheometer)**: Kinematic velocity observations within the cavity are supplied together with roller boundary values. The network reconstructs the unobserved latent stress and pressure fields while simultaneously discovering the governing physical parameters ($\lambda, \mu_p, \mu_s, \alpha, \varepsilon$) through physics-constrained optimization.

Table 4.1 summarizes the role of each field variable and parameter across both modes.

Table 4.1: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	**Symbol**	**Forward Problem**	**Inverse Problem**
Velocity Field	$\boldsymbol{u} = (u, v)$	Observed (via ψ)	Observed (via ψ)
Pressure Field	p	Latent unknown (PDE-resolved)	Latent unknown (PDE-resolved)
Polymeric Extra-Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Latent unknown (PDE-resolved)	Latent unknown (PDE-resolved)
Relaxation Time	λ (Wi)	Prescribed known	Unknown parameter
Polymeric Viscosity	μ_p	Prescribed known	Unknown parameter
Solvent Viscosity	μ_s	Prescribed known	Unknown parameter
Giesekus Mobility	α	Prescribed known	Unknown parameter
PTT Extensibility	ε	Prescribed known	Unknown parameter

4.2 Reference CFD Dataset

4.2.1 Finite Element Setup

Reference numerical solutions were generated using COMSOL Multiphysics [27], solving steady incompressible viscoelastic flow on unstructured triangular meshes.

The governing momentum and continuity equations were discretized using Taylor-Hood mixed finite elements (P_2–P_1), combining quadratic Lagrange shape functions for velocity components with linear shape functions for pressure to satisfy the inf-sup (Ladyzhenskaya-Babuška-Brezzi) stability condition. Viscoelastic extra-stress tensor components were interpolated using quadratic elements. To overcome the high-Weissenberg-number problem (HWNP) and suppress spurious node-to-node oscillations in convective stress transport at $Wi = 1.667$, numerical stabilization was enforced using Streamline Upwind Petrov-Galerkin (SUPG) streamline diffusion coupled with the Discrete Elastic-Viscous Split Stress (DEVSS) formulation. To ensure that reference fields are free from discretization artifacts, we evaluated spatial mesh convergence.

The canonical linear Oldroyd-B benchmark at $\lambda = 1.0\,\mathrm{s}$ ($Wi = 1.667$) was selected as the representative test case. Because linear Oldroyd-B lacks quadratic or exponential stress saturation mechanisms, normal stresses grow steeply near the central stagnation point, providing a conservative worst-case baseline for spatial resolution. Mesh independence was verified on this representative configuration and confirmed across all other fluid models and regimes investigated.

We evaluated six progressive grid resolutions:

- **M5k (Coarse Grid)**: 5,086 nodes, capturing macroscopic cavity recirculation.
- **M12k (Light Grid)**: 12,760 nodes.
- **M29k (Intermediate Grid)**: 29,392 nodes.
- **M52k (Fine Grid)**: 52,648 nodes.
- **M88k (Very Fine Grid)**: 88,804 nodes.
- **M125k (Dense Baseline)**: 125,456 nodes, resolving sub-millimeter viscoelastic stress layers within the narrow roller gaps.

4.2.2 Mesh Convergence Study

To quantitatively verify local kinematic and stress fields across mesh densities, three diagnostic cutlines are established (Figure 4.1):

1. **Horizontal Cutline** ($y = 0$, $-L/2 \leq x \leq L/2$): Outflow elongational axis passing through the central hyperbolic stagnation point $(0, 0)$, along which tensile normal stress tails τ_{xx} develop.

2. **Vertical Cutline** ($x = 0$, $-L/2 \leq y \leq L/2$): Compressive inflow axis capturing fluid deceleration toward the stagnation zone.

3. **Oblique Diagonal Cutline** ($y = x$): Connects opposite cavity corners across roller nips, capturing the transition from the extensional core into high-shear roller boundary layers.

Figure 4.2 displays spatial profiles of velocity and extra-stress tensor components $(u, v, \tau_{xx}, \tau_{xy}, \tau_{yy})$ across the cutlines. Spatial velocity profiles between M52k, M88k, and M125k deviate by less than 0.05%, while peak shear stresses in roller gaps differ by under 0.3%, confirming that M125k is fully grid-independent and provides a reliable ground truth.

4.3 Network Architecture and Loss Formulation

4.3.1 Decoupled Three-Head Network

To satisfy mass conservation identically in two-dimensional steady incompressible flow ($\nabla \cdot \boldsymbol{u} = 0$), the velocity field is parameterized through a scalar stream function $\psi(x, y)$:

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{4.1}$$

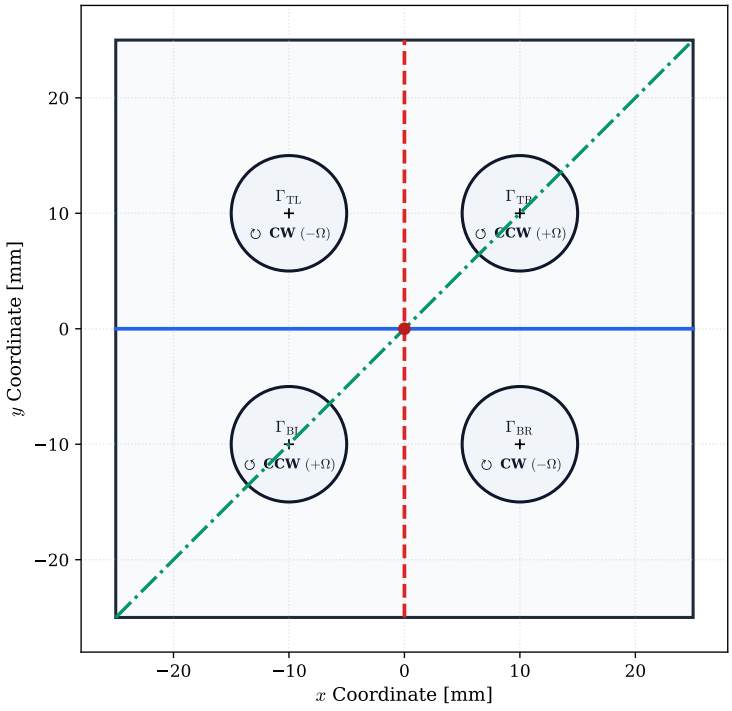


Figure 4.1: Schematic of the four-roll mill geometry with the four counter-rotating cylinders, central extensional stagnation point $(0, 0)$ (red circle), and three diagnostic verification cutlines: Horizontal ($y = 0$, outflow, blue solid), Vertical ($x = 0$, inflow, red dashed), and Oblique Diagonal ($y = x$, roller nips, green dash-dot).

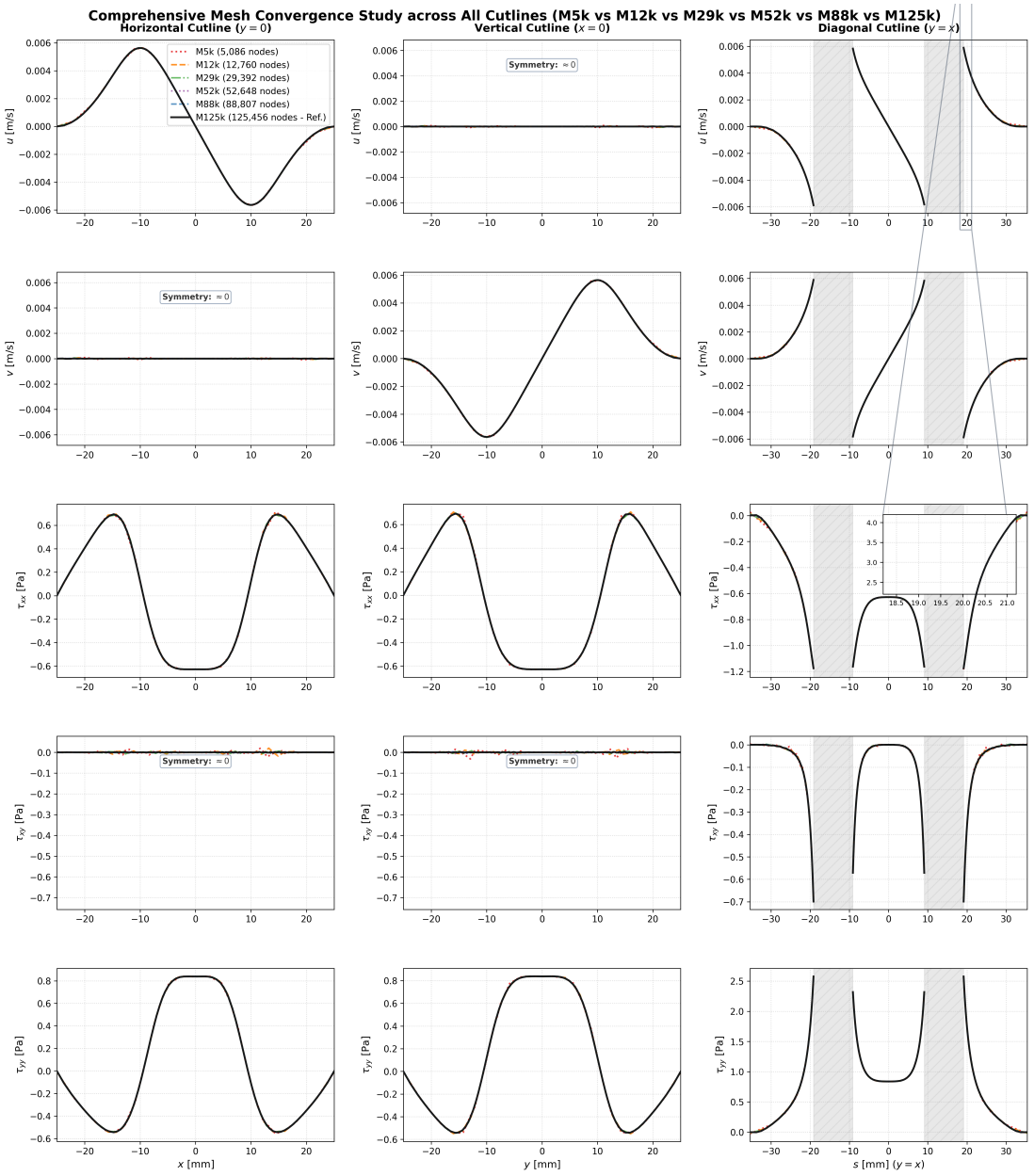


Figure 4.2: High-fidelity COMSOL mesh convergence matrix for the representative Oldroyd-B fluid ($\lambda = 1.0\,\mathrm{s}$) across six mesh densities (M5k to M125k) along the Horizontal ($y = 0$), Vertical ($x = 0$), and Oblique Diagonal ($y = x$) cutlines for velocity components (u, v) and stress tensor components $(\tau_{xx}, \tau_{xy}, \tau_{yy})$. Shaded gray vertical bands in the diagonal cutline panels indicate the solid interior occupied by the cylindrical rollers.

By defining kinematics through ψ, the incompressibility constraint is satisfied by construction, eliminating the continuity penalty term from the loss function.

As shown in Figure 4.3, the physical fields are predicted by three independent Multi-Layer Perceptron (MLP) branches rather than a single shared network:

$$(x, y) \in \mathbb{R}^2 \longrightarrow \begin{cases} \psi(x, y) & \text{(Stream function, yielding } u, v \text{ via AD)}, \\ p(x, y) & \text{(Hydrodynamic pressure)}, \\ \boldsymbol{\tau}(x, y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) & \text{(Extra-stress tensor)}. \end{cases} \tag{4.2}$$

Each branch contains 8 hidden layers with 128 neurons per layer, activated by the Sigmoid Linear Unit (SiLU$(z) = z/(1+e^{-z})$). Its smooth, infinitely differentiable profile provides well-behaved second- and third-order derivative graphs during automatic differentiation.

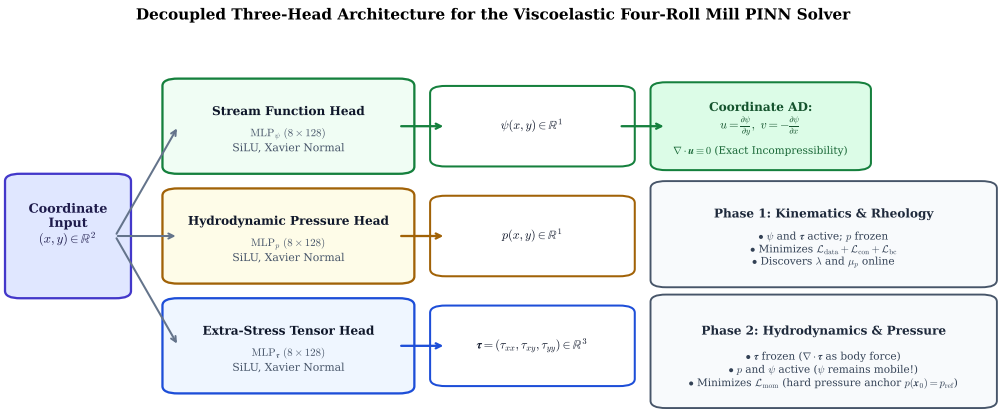


Figure 4.3: Decoupled three-head neural network architecture for the viscoelastic four-roll mill solver, showing coordinate inputs, separate MLP heads for stream function ψ, pressure p, and stress tensor $\boldsymbol{\tau}$, coordinate automatic differentiation, and the two-stage optimization schedule.

4.3.2 Weight Initialization

Hidden layers are initialized with the Xavier normal distribution, scaling weight variance inversely with layer width to maintain balanced gradient amplitudes across depth.

The final linear layers of the extra-stress network $\mathcal{N}_{\boldsymbol{\tau}}$ and pressure network $\mathcal{N}_p$ are initialized to zero ($\boldsymbol{W}^{(L)} = \mathbf{0}, \boldsymbol{b}^{(L)} = \mathbf{0}$). This choice provides two practical benefits:

1. **Quiescent Starting State**: At iteration zero, the model predicts zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$) and uniform pressure, reflecting a fluid initially at rest before deformation begins.

2. **Numerical Stability**: Random initial weights in the stress network produce high-frequency spatial fluctuations. Because constitutive models contain nonlinear terms ($\alpha\boldsymbol{\tau}^2$ in Giesekus) and convective terms ($\boldsymbol{u} \cdot \nabla\boldsymbol{\tau}$), uncalibrated stresses at step zero can generate extreme loss spikes ($> 10^6$) that disrupt the optimizer. Initializing output layers to zero avoids these early instabilities.

4.3.3 Governing Residuals and Scaling

Following the nondimensionalization introduced in Section 2.4, characteristic scales are $H_{\text{ref}} = R = 0.005\,\text{m}$ (roller radius) and $U_{\text{ref}} = u_{\text{roll}} = 8.333 \times 10^{-3}\,\text{m/s}$. In the inverse problem, fluid viscosities μ_s and μ_p are unknown a priori. To decouple the coordinate scaling from evolving parameter estimates, the solver sets a fixed reference viscosity scale $\eta_0 = 1.0\,\text{Pa} \cdot \text{s}$:

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0}, \quad Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0}, \quad Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}}. \tag{4.3}$$

Spatial coordinates are mapped to the unit box $[0, 1]^2$ using the cavity width

$H_{\text{coord}} = L = 0.05\,\text{m}$ to keep activation arguments well conditioned. Derivative operators with respect to physical space are scaled accordingly via Coordinate AD by $(H_{\text{ref}}/H_{\text{coord}})$.

In contrast to the coordinates, the physical output fields (p and stress components $\tau_{xx}, \tau_{xy}, \tau_{yy}$) oscillate symmetrically with both positive and negative values across the cavity. To keep raw network outputs in the balanced range $[-1, 1]$, each output head is scaled by its maximum absolute reference value $(p_{\text{scale}}, \boldsymbol{\tau}_{\text{scale}})$:

$$p(x, y) = p_{\text{scale}} \cdot \hat{p}(x, y), \qquad \boldsymbol{\tau}(x, y) = \boldsymbol{\tau}_{\text{scale}} \odot \hat{\boldsymbol{\tau}}(x, y), \tag{4.4}$$

where $\hat{p}, \hat{\boldsymbol{\tau}} \in [-1, 1]$ are the normalized network outputs. Hereafter, denoting the dimensionless reconstructed physical fields scaled by the characteristic reference quantities of Section 2.4.2 simply as $(u, v, p, \boldsymbol{\tau})$, the governing PDE residuals evaluated at interior collocation points $\boldsymbol{x}_m \in \Omega$ evaluate to:

1. **Linear Momentum Residuals** (f_u, f_v):

$$f_u = Re_{\text{scale}} \left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} \right) + \frac{\partial p}{\partial x} - \mu_s^* \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right), \tag{4.5}$$

$$f_v = Re_{\text{scale}} \left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} \right) + \frac{\partial p}{\partial y} - \mu_s^* \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right). \tag{4.6}$$

2. **Viscoelastic Constitutive Residuals** $(f_{\tau_{xx}}, f_{\tau_{xy}}, f_{\tau_{yy}})$:

$$f_{\tau_{xx}} = \mathcal{G}_{xx}(\boldsymbol{\tau}) + Wi\, \overset{\nabla}{\tau}_{xx} - 2\mu_p^* \frac{\partial u}{\partial x}, \tag{4.7}$$

$$f_{\tau_{yy}} = \mathcal{G}_{yy}(\boldsymbol{\tau}) + Wi\, \overset{\nabla}{\tau}_{yy} - 2\mu_p^* \frac{\partial v}{\partial y}, \tag{4.8}$$

$$f_{\tau_{xy}} = \mathcal{G}_{xy}(\boldsymbol{\tau}) + Wi\, \overset{\nabla}{\tau}_{xy} - \mu_p^* \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \tag{4.9}$$

where $\overset{\nabla}{\tau}_{ij}$ denotes the upper-convected derivative (Eq. (2.17)), and $\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ defines the constitutive model:

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B)}, \\ \boldsymbol{\tau} + \dfrac{\alpha Wi}{\mu_p^*}\boldsymbol{\tau}^2 & \text{(Giesekus)}, \\ \left(1 + \dfrac{\varepsilon Wi}{\mu_p^*}\mathrm{tr}(\boldsymbol{\tau})\right)\boldsymbol{\tau} & \text{(Linear PTT)}. \end{cases} \tag{4.10}$$

4.3.4 Composite Loss Function and Roller Stress Anchor

The composite objective function minimized during training is:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}}. \tag{4.11}$$

Because spatial coordinates ($[0,1]^2$), kinematic velocities ($\mathcal{O}(1)$ via U_{ref} and variance scaling), and stress/pressure outputs are systematically scaled to dimensionless order-unity values, the individual loss components remain naturally balanced without requiring stiff empirical weighting. Accordingly, the baseline weights are set uniformly to unity: $w_{\text{data}} = 1.0$, $w_{\text{bc}} = 1.0$, and $w_{\text{con}} = 1.0$. The momentum balance weight follows the staged training schedule ($w_{\text{mom}} = 0.0$ in Phase 1, $w_{\text{mom}} = 1.0$ in Phase 2), while $w_{\text{roll_stress}} = 1.0$ is selectively activated during inverse parameter discovery and disabled ($w_{\text{roll_stress}} = 0$) during direct field reconstruction.

The individual loss terms are defined as follows:

- **Kinematic Data Loss** ($\mathcal{L}_{\text{data}}$): Quantifies the mean squared error against observed dimensionless velocity data:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}}\sum_{i=1}^{N_{\text{data}}}\left[\frac{(\hat{u}(\boldsymbol{x}_i) - u_i^{\text{data}})^2}{\sigma_u^2} + \frac{(\hat{v}(\boldsymbol{x}_i) - v_i^{\text{data}})^2}{\sigma_v^2}\right], \tag{4.12}$$

where $\hat{u}(\boldsymbol{x}) = \frac{\partial \hat{\psi}}{\partial y}$, $\hat{v}(\boldsymbol{x}) = -\frac{\partial \hat{\psi}}{\partial x}$, and σ_u^2, σ_v^2 are empirical dimensionless field variances used to scale horizontal and vertical kinematic errors to comparable magnitude.

- **Kinematic Boundary Loss** ($\mathcal{L}_{\text{bc}}$): Enforces no-slip conditions on outer cavity walls and prescribed tangential velocities on the rotating cylinder surfaces in dimensionless variables:

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_j) - \boldsymbol{u}^{\text{wall/roll}}(\boldsymbol{x}_j) \right\|_2^2 . \tag{4.13}$$

- **Constitutive Residual Loss** ($\mathcal{L}_{\text{con}}$): Evaluates the mean squared residual of the rheological equations over collocation points:

$$\mathcal{L}_{\text{con}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \left[f_{\tau_{xx}}(\boldsymbol{x}_m)^2 + f_{\tau_{xy}}(\boldsymbol{x}_m)^2 + f_{\tau_{yy}}(\boldsymbol{x}_m)^2 \right] . \tag{4.14}$$

- **Momentum Residual Loss** ($\mathcal{L}_{\text{mom}}$): Evaluates linear momentum balance over collocation points:

$$\mathcal{L}_{\text{mom}} = \frac{1}{N_{\text{col}}} \sum_{m=1}^{N_{\text{col}}} \left[f_u(\boldsymbol{x}_m)^2 + f_v(\boldsymbol{x}_m)^2 \right] . \tag{4.15}$$

- **Pressure Gauge Fixing**: In incompressible flow, pressure is determined only up to an additive constant. Rather than adding a soft penalty term to the loss, our implementation anchors pressure algebraically in the network forward pass:

$$p(\boldsymbol{x}) = p_{\text{scale}} \left(\mathcal{N}_p(\boldsymbol{x}) - \mathcal{N}_p(\boldsymbol{x}_0) \right) + p_{\text{ref}}, \tag{4.16}$$

where $\boldsymbol{x}_0$ is a fixed reference point on the cavity wall. This guarantees $p(\boldsymbol{x}_0) \equiv p_{\text{ref}}$ identically, fixing the hydrostatic gauge without requiring an extra penalty weight.

Roller Stress Anchor

The roller stress supervision term,

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \left\| \boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}} \right\|_2^2, \tag{4.17}$$

plays a specific physical role in the inverse problem. In creeping flow ($Re \ll 1$), multiplying viscosities (μ_s, μ_p), pressure p, and extra-stress $\boldsymbol{\tau}$ by an arbitrary positive constant $C > 0$ leaves the velocity field $\boldsymbol{u}(x, y)$ unchanged:

$$-\nabla(Cp) + (C\mu_s)\nabla^2\boldsymbol{u} + \nabla \cdot (C\boldsymbol{\tau}) = C\left[-\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau}\right] = \mathbf{0}. \tag{4.18}$$

Because velocity data alone cannot resolve this multiplicative ambiguity in Pa · s, prescribing $\boldsymbol{\tau}$ exclusively along the roller boundaries provides the necessary scale anchor, removing the scaling degeneracy without requiring interior stress observations. In the forward problem, where fluid viscosities are known a priori, this term is disabled ($w_{\text{roll_stress}} = 0$).

From an experimental rheometry viewpoint, supervising stress exclusively along the roller cylinders Γ_{roll} is both practically feasible and physically realistic. In physical four-roll mill apparatuses [6, 7], the rotating cylinders are mounted on instrumented drive shafts equipped with calibrated torque transducers. Measuring the total hydrodynamic torque $\mathcal{T}_{\text{roll}} = R \oint_{\Gamma_{\text{roll}}} \tau_{r\theta}\, \mathrm{d}s$ provides direct experimental access to the cylinder surface traction. Probing localized extra-stress inside the internal fluid bulk would require complex full-field optical polarimetry or flow birefringence, which are notoriously delicate to calibrate quantitatively. Anchoring stress exclusively at the roller boundary therefore replicates the actual information available in a physical rheometrical experiment without unphysically assuming access to interior stress fields.

4.4 Training Strategy

4.4.1 Two-Stage Physical Decoupling

Optimizing momentum and constitutive equations together from scratch often leads to numerical instability. The hyperbolic stress transport terms produce gradient magnitudes that overwhelm the elliptic momentum balance, causing the optimizer to stall or produce unphysical stress concentrations.

To avoid this imbalance, we split training into two sequential stages:

1. **Phase 1: Kinematics and Rheology (Stress Network Training)**: The momentum balance is turned off ($w_{\text{mom}} = 0$) and the pressure network $\mathcal{N}_p$ remains frozen. The stream function ψ and extra-stress $\boldsymbol{\tau}$ are trained jointly by minimizing kinematic data matching alongside constitutive residuals:

$$\mathcal{L}_{\text{Phase 1}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}}. \tag{4.19}$$

   In inverse mode, unknown fluid parameters (λ, μ_p, α) are identified during this phase, isolating constitutive dynamics from hydrodynamic pressure interactions.

2. **Phase 2: Hydrodynamics and Pressure Recovery (Momentum Training)**: The extra-stress network $\mathcal{N}_{\boldsymbol{\tau}}$ is frozen, and its spatial divergence $\nabla \cdot \boldsymbol{\tau}$ acts as a fixed body force. The momentum loss is activated ($w_{\text{mom}} = 1.0$) while pressure is anchored algebraically at $\boldsymbol{x}_0$ (Eq. (4.16)):

$$\mathcal{L}_{\text{Phase 2}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{mom}}\mathcal{L}_{\text{mom}}. \tag{4.20}$$

Crucially, the stream function network $\mathcal{N}_\psi$ must remain trainable during

Phase 2. This formulation resolves two fundamental limitations observed in prior viscoelastic PINN architectures, notably ViscoelasticNet [9]. First, ViscoelasticNet predicts velocity components (u, v) with independent network heads and penalizes divergence through a soft penalty loss $\mathcal{L}_{\text{div}} = \|\nabla \cdot \boldsymbol{u}\|_2^2$, resulting in competing gradients and only approximate mass conservation. In contrast, our stream-function formulation $\psi(x, y)$ identically satisfies $\nabla \cdot \boldsymbol{u} \equiv 0$ everywhere to machine precision.

Second, ViscoelasticNet's staged training freezes the velocity field during pressure and momentum estimation. In closed recirculating cavities, this presents an insurmountable physical obstacle. Pressure produces purely irrotational forces ($\nabla \times \nabla p \equiv \mathbf{0}$). Taking the curl of the momentum equations eliminates pressure completely:

$$\nabla \times \nabla p \equiv \mathbf{0} \implies \nabla \times \left(\mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau} - \rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u})\right) = \mathbf{0}. \tag{4.21}$$

Pressure can push or expand fluid elements, but it cannot balance rotational forces (vorticity). If ψ were frozen after Phase 1, any residual rotational mismatch between viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ and stress divergence $\nabla \cdot \boldsymbol{\tau}$ could never be compensated by pressure, no matter what field $p(x, y)$ the network predicts. Keeping ψ trainable allows the velocity field to make minor sub-percent adjustments that reconcile the rotational part of the momentum balance, leaving the irrotational part to be cleanly balanced by the pressure gradient ∇p.

Table 4.2 summarizes the training stages, active heads, and loss weights.

Table 4.2: Summary of the decoupled two-stage training strategy.

Stage	**Trainable Heads**	**Frozen Heads**	**Active Objectives**
Phase 1	ψ, $\boldsymbol{\tau}$	p	$\mathcal{L}_{\text{data}} + \mathcal{L}_{\text{bc}} + \mathcal{L}_{\text{con}} + \mathcal{L}_{\text{roll_stress}}$
Phase 2	p, ψ (mobile)	$\boldsymbol{\tau}$	$\mathcal{L}_{\text{data}} + \mathcal{L}_{\text{bc}} + \mathcal{L}_{\text{mom}}$ (Hard p-anchor)

4.4.2 Dual-Precision Optimization

Optimization proceeds in two sequential precision tiers:

1. **Tier 1: Fast Exploration via Adam (FP32)**: In early iterations, the loss landscape contains local minima and steep gradients. The Adam optimizer runs in single precision (FP32) to take advantage of GPU throughput. A cosine annealing schedule ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$) guides parameter values into the attraction basin of the physical solution.

2. **Tier 2: Fine-Tuning with L-BFGS in Double Precision (FP64)**: To resolve higher-order spatial derivatives without round-off errors and eliminate stochastic fluctuations, the model switches to double precision (`torch.float64`). L-BFGS with a Strong Wolfe line search and history size $m = 50$ provides superlinear convergence, driving PDE residuals down to tight numerical tolerances.

4.4.3 Transfer Learning Protocol

Training PINNs from scratch for every fluid regime requires substantial computational budgets (50,000 iterations) and risks local minima at high elasticity. We establish a Transfer Learning protocol:

1. Pre-train the base network on a moderate baseline case ($\lambda = 0.05\,\mathrm{s}, Wi = 0.08$).

2. Transfer pre-trained weights as initializations for target regimes characterized by high elasticity ($\lambda = 0.70\,\mathrm{s}, Wi = 1.17$) or strong non-linearity ($\alpha = 0.50$).

3. Fine-tune with halved iteration budgets (25,000 iterations), achieving a 50% reduction in computational duration while preventing high-Wi optimization divergence.

Table 4.3 details the hyperparameters and iteration budgets for both precision tiers and initialization strategies.

Table 4.3: Optimization hyperparameters, precision schedule, and transfer learning settings.

Tier	Optimizer	Precision	Iterations (Scratch / TL)	Learning Rate / Line Search
Tier 1	Adam	FP32	40,000 / 20,000	Cosine Annealing ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$)
Tier 2	L-BFGS	FP64	10,000 / 5,000	Strong Wolfe ($\alpha_0 = 1.0$, $m = 50$)

All network training and optimization routines were executed locally on a single workstation equipped with an NVIDIA GeForce RTX 3060 GPU (12 GB VRAM, Ampere architecture, CUDA 12.2) and an Intel Core i7 processor. On the dense M125k grid (125,456 nodes), a complete ex-novo training run of 50,000 iterations takes approximately 78 minutes for Phase 1 (40k Adam iterations in FP32 require ∼46 min, followed by 10k L-BFGS iterations in FP64 taking ∼32 min) and 28 minutes for Phase 2 (Adam plus L-BFGS refinement), totaling approximately 1.75 hours (∼106 minutes). On coarser meshes such as M12k, Phase 1 completes in under 30 minutes. Applying Transfer Learning halves the required iterations to 25,000, reducing Phase 1 wall-clock time on M125k to approximately 39 minutes while preventing numerical divergence at high Weissenberg numbers.

4.5 Direct Benchmark: Latent Stress and Pressure Reconstruction

We evaluate the PINN framework on the canonical Oldroyd-B benchmark ($Wi = 0.0833, Re = 0.0417, \lambda = 0.05\,\mathrm{s}, \mu_p = 0.90\,\mathrm{Pa}\cdot\mathrm{s}, \mu_s = 0.10\,\mathrm{Pa}\cdot\mathrm{s}$) under known rheology. Planar velocity observations supervise the stream function $\hat{\psi}$, while the network autonomously reconstructs the unmeasured latent extra-stress tensor $\hat{\boldsymbol{\tau}}(x, y)$ and hydrodynamic pressure $\hat{p}(x, y)$ strictly through the momentum and constitutive residuals.

4.5.1 Velocity and Stress Reconstruction

Field accuracy is quantified using the relative global L_2 error norm evaluated over all $N = 125{,}456$ nodes:

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2} = \frac{\sqrt{\sum_{i=1}^{N} |\hat{\phi}(\boldsymbol{x}_i) - \phi^{\text{ref}}(\boldsymbol{x}_i)|^2}}{\sqrt{\sum_{i=1}^{N} |\phi^{\text{ref}}(\boldsymbol{x}_i)|^2}}. \tag{4.22}$$

Complementing the integral L_2 norm, local spatial error distributions in contour maps are reported via the range-relative error:

$$\epsilon_{\text{range}}(\boldsymbol{x}) = \frac{|\hat{\phi}(\boldsymbol{x}) - \phi^{\text{ref}}(\boldsymbol{x})|}{\max_\Omega(\phi^{\text{ref}}) - \min_\Omega(\phi^{\text{ref}})} \times 100\%, \tag{4.23}$$

which normalizes point-wise discrepancies against the full dynamic span of the field.

Table 4.4 summarizes the validation metrics on the M125k grid.

Table 4.4: Global hydrodynamic field accuracy for the direct Oldroyd-B benchmark ($Wi = 0.0833, Re = 0.0417$) on the M125k evaluation grid.

Field Variable	**Symbol**	**Relative L_2 Error**	**Physical Assessment**
Horizontal Velocity	u	**0.98**%	Outflow jets and four-fold symmetry resolved
Vertical Velocity	v	**0.93**%	Exact incompressibility satisfied via ψ
Shear Extra-Stress	τ_{xy}	**1.21**%	Localized peak shear in roller nips resolved
Axial Normal Stress	τ_{xx}	**1.80**%	Elongational stretching along extension axis
Transverse Normal Stress	τ_{yy}	**1.65**%	Compressive stress along inflow streamlines
Hydrodynamic Pressure	p (Calibrated)	**3.42**%	Physical gradient ∇p balances stress divergence

Figure 4.4 compares the predicted dimensionless velocity components ($u^* = u/U_{\text{ref}}, v^* = v/U_{\text{ref}}$) against the COMSOL reference (L_2 errors of 0.98% for u and 0.93% for v, with local range-relative errors strictly bounded below 3.5%). The stream-function parameterization maintains the four-fold flow symmetry and accurately locates the central hyperbolic stagnation point.

Figure 4.5 shows the corresponding dimensionless extra-stress components ($\boldsymbol{\tau}^* = \boldsymbol{\tau}/\tau_{\text{ref}}$), with relative L_2 errors of 1.21% on shear stress τ^*_{xy} and 1.80% on normal stress τ^*_{xx}. Stresses are highest in the narrow roller gaps, where intense ve-

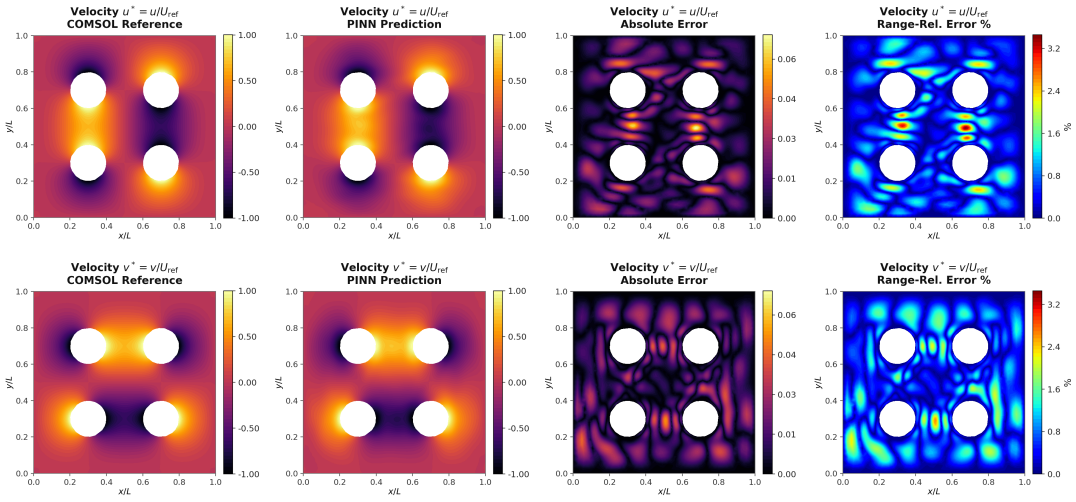


Figure 4.4: Direct problem kinematic validation: comparison of dimensionless horizontal ($u^* = u/U_{\text{ref}}$) and vertical ($v^* = v/U_{\text{ref}}$) velocity fields on normalized coordinates $[0, 1]^2$ across COMSOL reference, PINN prediction, absolute error, and range-relative error on the M125k mesh.

locity shearing occurs, and the network reproduces these localized peaks without visible numerical oscillations.

4.5.2 Pressure Recovery and Gauge Calibration

In incompressible flow, absolute pressure p is determined only up to an arbitrary additive constant C, because only its spatial gradient ∇p enters the momentum equations. In our architecture, the hydrostatic gauge is set by the algebraic anchor at a boundary wall point $\boldsymbol{x}_0$ ($p(\boldsymbol{x}_0) = p_{\text{ref}}$). However, discrete numerical solvers often use different gauge conventions; for instance, COMSOL fixes the average pressure across the entire domain to zero.

Evaluating raw predictions against COMSOL produces an apparent offset error (47.4%) that is purely hydrostatic. To assess the physical agreement of the pressure gradient, we align the domain-mean gauge in post-processing by

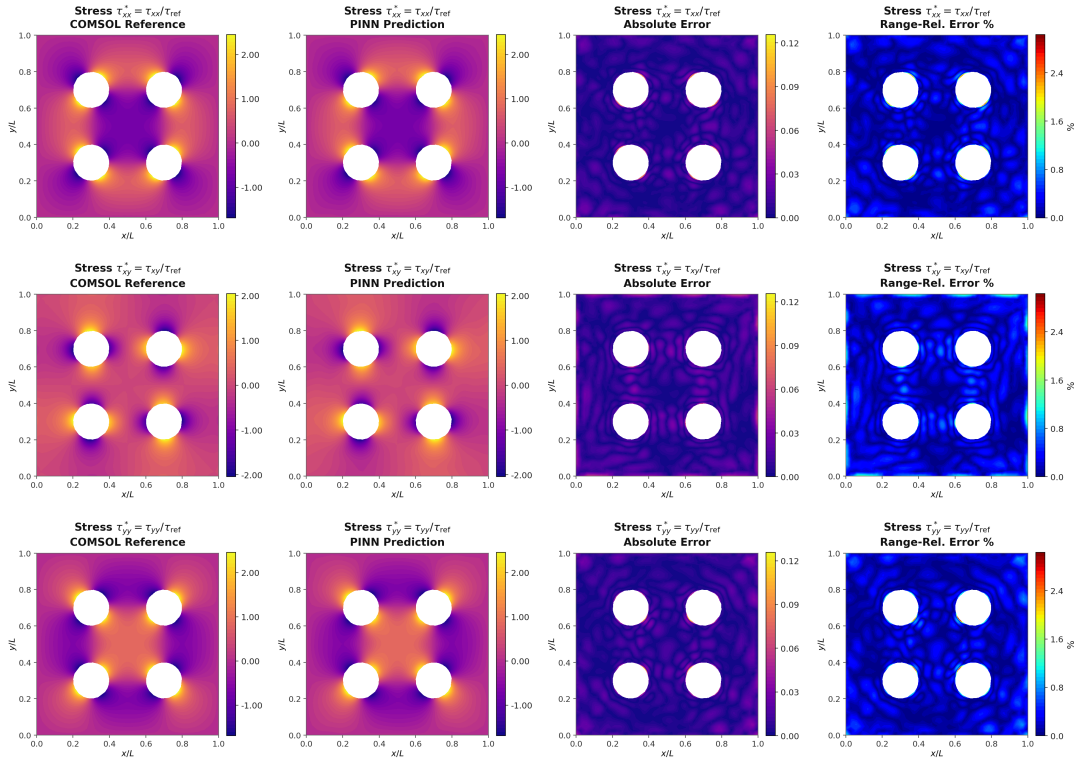


Figure 4.5: Direct problem viscoelastic extra-stress tensor validation: dimensionless COMSOL ground truth ($\boldsymbol{\tau}^* = \boldsymbol{\tau}/\tau_{\text{ref}}$), PINN predictions, absolute errors, and range-relative errors for normal components (τ^*_{xx}, τ^*_{yy}) and shear component τ^*_{xy} on normalized coordinates $[0, 1]^2$.

subtracting the mean shift:

$$p_{\text{cal}}(\boldsymbol{x}) = \hat{p}(\boldsymbol{x}) - \frac{1}{|\Omega|}\int_{\Omega}\left(\hat{p}(\boldsymbol{x}) - p^{\text{ref}}(\boldsymbol{x})\right)\mathrm{d}\Omega. \tag{4.24}$$

Because this shift is uniform, it leaves the physical gradient unchanged ($\nabla p_{\text{cal}} \equiv \nabla\hat{p}$). As shown in Figure 4.6, the calibrated pressure matches COMSOL with a relative L_2 error of 3.42% (range-normalized error $< 3.5\%$), confirming that the reconstructed gradient balances viscous and elastic forces accurately.

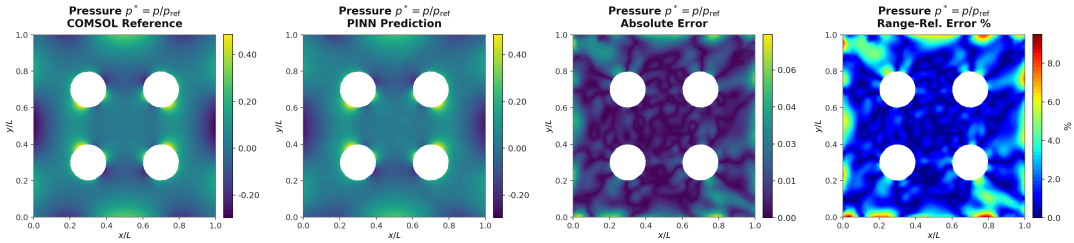


Figure 4.6: Hydrodynamic pressure field reconstruction: dimensionless COMSOL ground truth ($p^* = p/p_{\text{ref}}$), PINN prediction, absolute error, and range-relative error distributions on normalized coordinates $[0, 1]^2$ (L_2 error of 3.42% after mean gauge calibration).

4.6 Inverse Parameter Discovery

In inverse discovery mode, the PINN operates as a virtual rheometer: no internal stress measurements are supplied. The network assimilates domain velocity data, solid boundary kinematic conditions, and the roller stress anchor ($\mathcal{L}_{\text{roll_stress}}$). Internal stresses and pressure remain unobserved, inferred simultaneously alongside unknown fluid parameters.

4.6.1 Oldroyd-B and Giesekus Benchmarks

We evaluate four canonical benchmarks spanning linear Hookean and non-linear shear-thinning response:

- **Benchmark 1 (Oldroyd-B from Scratch)**: Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations starting from blind offset guesses ($\lambda_0 = 0.0535\,\mathrm{s}$ and $\mu_p^0 = 0.963\,\mathrm{Pa}\cdot\mathrm{s}$, both +7% from truth).

- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning)**: Elevated elasticity ($Wi = 1.17, \lambda = 0.70\,\mathrm{s}$), fine-tuned via TL for 25,000 iterations.

- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch)**: Non-linear mobility ($\alpha^{\mathrm{true}} = 0.35$), initialized from a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.

- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning)**: Maximum physical mobility ($\alpha^{\mathrm{true}} = 0.50, \lambda = 0.30\,\mathrm{s}, \mu_p = 0.05\,\mathrm{Pa}\cdot\mathrm{s}$), fine-tuned via TL for 25,000 iterations.

Table 4.5 summarizes the discovery accuracy and kinematic errors across these benchmarks.

Table 4.5: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Model	Strategy	Target Parameter	PINN Estimation	Error	$L_2(u,v)$
1	**Oldroyd-B**	Ex-Novo	$\lambda = 0.050\,\mathrm{s}$	$\mathbf{0.0502}\,\mathrm{s}$	$+\mathbf{0.41}\%$	0.04%
	($Wi = 0.08$)	(50k iters)	$\mu_p = 0.900\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.9049}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{0.54}\%$	
2	**Oldroyd-B**	Transfer Learning	$\lambda = 0.700\,\mathrm{s}$	$\mathbf{0.7089}\,\mathrm{s}$	$+\mathbf{1.27}\%$	0.33%
	($Wi = 1.17$)	(25k iters, -50%)	$\mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.5019}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{0.38}\%$	
3	**Giesekus**	Ex-Novo (Blind)	$\alpha = 0.350$	$\mathbf{0.3298}$	$-\mathbf{5.76}\%$	
	($\alpha_0 = 0.25$)	(50k iters)	$\lambda = 0.100\,\mathrm{s}$	$\mathbf{0.1032}\,\mathrm{s}$	$+\mathbf{3.16}\%$	0.46%
			$\mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.5307}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{6.14}\%$	
4	**Giesekus**	Transfer Learning	$\alpha = 0.500$	$\mathbf{0.4665}$	$-\mathbf{6.70}\%$	
	(Extreme α)	(25k iters, -50%)	$\lambda = 0.300\,\mathrm{s}$	$\mathbf{0.3077}\,\mathrm{s}$	$+\mathbf{2.57}\%$	0.52%
			$\mu_p = 0.050\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.0522}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{4.40}\%$	

Parameter estimation errors for Oldroyd-B remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For Giesekus, the solver simultaneously inverts three coupled parameters (α, λ, μ_p), identifying mobility α within 6.7% error even from a blind initialization ($\alpha_0 = 0.25$ vs 0.35).

Figure 4.7 illustrates parameter trajectories and objective function evolution during staged Adam (FP32, initial 40k epochs) and L-BFGS (FP64, final 10k iterations) optimization on the M125k dataset. In Figure 4.7(left), the normalized parameter trajectories $\lambda/\lambda_{\mathrm{GT}}$ and $\mu_p/\mu_{p,\mathrm{GT}}$ initially rise from 1.07 to an overshoot peak near 1.28 during the first 2,500 Adam iterations before reversing course and converging monotonically toward 1.00. This transient behavior has a direct physical cause: at iteration zero, the stress network $\mathcal{N}_{\boldsymbol{\tau}}$ is initialized near zero ($\boldsymbol{\tau} \approx \mathbf{0}$). In the absence of a structured stress field, the constitutive residual gradients momentarily push parameter estimates upward to balance the prescribed deformation rate. Once the stress head establishes the localized boundary layers and shear bands (around epoch 3,000), the optimizer reorients toward the physical minimum, guiding parameters into the physical neighborhood. Upon switching to the second-order L-BFGS optimizer at iteration 40k, the kinematic data loss drops sharply by an order of magnitude (from 10^{-6} to 1.81×10^{-7}), while the constitutive loss settles monotonically to $\mathcal{L}_{\mathrm{con}} = 6.13 \times 10^{-6}$, providing smooth asymptotic settling without numerical oscillations.

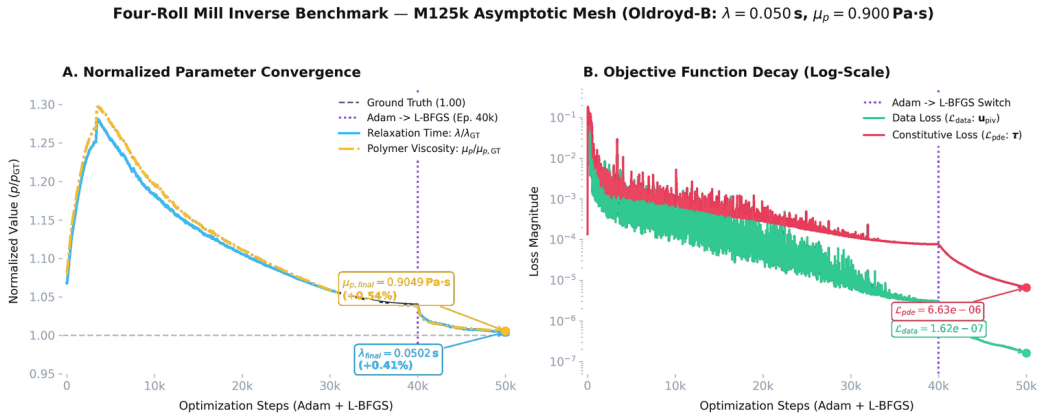


Figure 4.7: Parameter convergence trajectories (λ and μ_p, left) and staged objective loss decay ($\mathcal{L}_{\mathrm{data}}$ and $\mathcal{L}_{\mathrm{con}}$, right) during Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

4.6.2 Transfer Learning Performance

The benchmarks in Table 4.5 demonstrate two practical benefits of transfer learning:

1. **Computational Acceleration**: Warm-starting from a pre-trained baseline reduces required iterations by half (25,000 vs. 50,000), halving training time.

2. **Optimization Regularization**: Initializing weights from a lower-Wi solution places the optimization in a physically reasonable neighborhood, avoiding the poor local minima that can occur when training from scratch at higher elasticity ($Wi = 1.17$) or high mobility ($\alpha = 0.50$).

4.6.3 Collocation Density Sensitivity

To examine how collocation sampling density affects PINN parameter discovery, we conducted a systematic grid study across five resolutions from 5k to 125k nodes (Table 4.6 and Figure 4.8) on an Oldroyd-B case ($\lambda = 0.100\,\mathrm{s}, \mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$).

Table 4.6: Collocation density sensitivity and parameter estimation accuracy across five resolutions ($\lambda^{\mathrm{true}} = 0.100\,\mathrm{s}, \mu_p^{\mathrm{true}} = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$).

Mesh Resolution	**Evaluation Nodes**	λ_{est} [s]	λ **Error**	$\mu_{p,\mathrm{est}}$ [Pa·s]	μ_p **Error**	**Velocity** L_2
5k (Coarse)	**5,086**	**0.1008**	**+0.79%**	**0.5103**	**+2.06%**	**0.15%**
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,392	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,648	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

Errors remain within a narrow band (0.8%–8.8%) across a 25-fold increase in point density, and accuracy does not improve monotonically: the 5k dataset achieves the lowest error (+0.79% on λ), while denser sets stabilize around 6–8%.

The discrepancy is explained by comparing with Benchmark 1 ($\lambda = 0.05\,\mathrm{s}$ on M125k, error +0.41%). Doubling λ to 0.10 s doubles the local Weissenberg number and steepens boundary layer stress gradients. On the dense M125k grid,

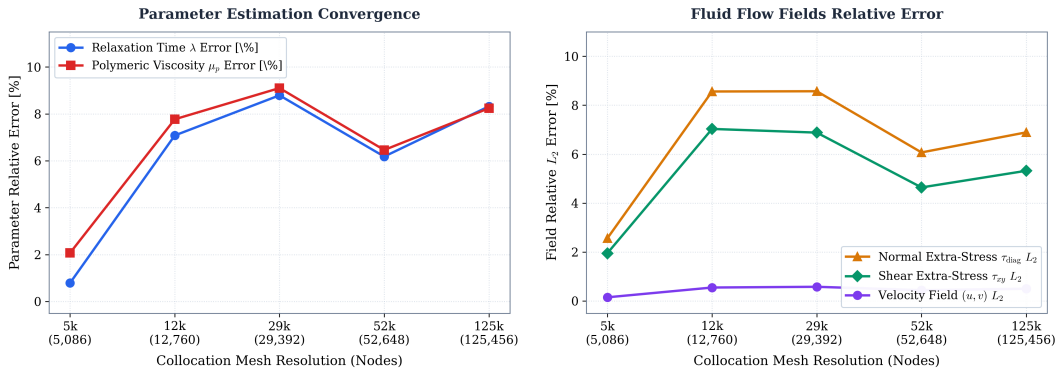


Figure 4.8: Collocation density study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across resolutions from 5,086 to 125,456 collocation points.

thousands of collocation points cluster in these high-curvature roller layers, shifting the gradient balance of the composite loss beyond what the fixed network ($\sim 1.5 \times 10^5$ parameters) can resolve without per-grid hyperparameter retuning. The coarser 5k grid distributes points more evenly across the cavity, which happens to favor the optimizer. In all cases, velocity kinematics remain consistently reconstructed ($L_2 \leq 0.58\%$).

4.7 Identifiability Boundaries and Invariance Principles

4.7.1 Elastic Shear Modulus Invariance

Across all inverse benchmarks, a consistent physical result emerges: while individual fluid parameters can be sensitive to boundary conditioning, the **composite Elastic Shear Modulus** $G = \mu_p/\lambda$ is identified with high precision across every model tested (Table 4.7).

$$G = \frac{\mu_p}{\lambda}. \tag{4.25}$$

In polymer kinetic theory, $G = nk_BT$ represents the macroscopic plateau modulus, measuring the reversible elastic energy storage per unit volume. The physical reason for its robust recovery becomes clear when dividing the general constitutive equation by the relaxation time λ:

$$\frac{1}{\lambda}\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) + \overset{\nabla}{\boldsymbol{\tau}} = 2\left(\frac{\mu_p}{\lambda}\right)\boldsymbol{D} = 2G\boldsymbol{D}. \tag{4.26}$$

In our inverse formulation, the velocity field $\boldsymbol{u}(x, y)$ is supervised by kinematic observations. Consequently, the rate-of-strain tensor $\boldsymbol{D}$ and the convective terms in $\overset{\nabla}{\boldsymbol{\tau}}$ are fixed by the data. The source term coupling kinematics to stress generation is $2G\boldsymbol{D}$. Neither G, nor $\boldsymbol{D}$, nor $\overset{\nabla}{\boldsymbol{\tau}}$ depends on λ individually. Any error in G directly violates the balance between prescribed deformation rate and convective stress transport across the entire cavity.

By contrast, the relaxation term $\frac{1}{\lambda}\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ acts as a secondary dissipative mechanism. In recirculating regions where local deformation and convection dominate, variations along the hyperbola $\mu_p = G\lambda$ alter only this weaker relaxation contribution. The loss function $\mathcal{L}_{\text{con}}(\lambda, \mu_p)$ therefore forms a steep canyon whose valley floor follows the line $\mu_p = G\lambda$. The optimizer rapidly converges onto this valley, locking onto G, while gradients along the valley remain comparatively flat.

Table 4.7: Identification of the macroscopic elastic shear modulus $G = \mu_p/\lambda$ across inverse benchmarks.

Benchmark Scenario	G^{true} [Pa]	G_{est} [Pa]	**Relative Error**	**Individual Errors** (λ, μ_p)
B1: Oldroyd-B ($Wi = 0.08$)	18.000	18.026	$+\mathbf{0.14}\%$	$+0.41\%, +0.54\%$
B2: Oldroyd-B ($Wi = 1.17$)	0.714	0.708	$-\mathbf{0.88}\%$	$+1.27\%, +0.38\%$
B3: Giesekus ($\alpha = 0.35$, blind)	5.000	5.142	$+\mathbf{2.85}\%$	$+3.16\%, +6.14\%$
B4: Giesekus ($\alpha = 0.50$, extreme)	0.167	0.170	$+\mathbf{1.79}\%$	$+2.57\%, +4.40\%$
PTT Case 1 ($\lambda = 1.0, \varepsilon = 0.3$)	0.500	0.506	$+\mathbf{1.19}\%$	$-57.89\%, -57.44\%$
PTT Case 2 ($\lambda = 0.7, \varepsilon = 0.5$)	1.357	1.367	$+\mathbf{0.70}\%$	$-61.43\%, -61.19\%$

As documented in Table 4.7, G_{est} is recovered within 3% across all four benchmarks. Most notably, in the two PTT cases where λ and μ_p individually collapse by approximately 58–61%, both parameters degrade by the exact same propor-

tion, preserving their ratio G within 1.2%.

From an optimization and loss-landscape perspective, this structural invariance is governed by four theoretical characteristics:

1. **Loss Hessian Anisotropy**: The local Hessian $\boldsymbol{H} = \nabla^2_{(\ln\lambda, \ln\mu_p)}\mathcal{L}_{\text{con}}$ exhibits ill-conditioning ($\kappa_1/\kappa_2 \gg 10^2$), with the stiff eigenvector oriented orthogonal to the line $\ln\mu_p - \ln\lambda = \text{const}$. Deviations from G are heavily penalized, whereas displacements along the G-preserving valley incur negligible gradient cost.

2. **Residual Canyon Geometry**: Evaluating the residual $\mathcal{L}_{\text{con}}(\lambda, \mu_p)$ analytically reveals an elongated parabolic canyon aligned with $\mu_p = G\lambda$, leaving individual parameters under-constrained under purely kinematic velocity observations.

3. **Initial Guess Orthogonal Projection**: Initializing the inverse solver from pairs (λ_0, μ_p^0) that share the correct initial ratio $G_0 = G_{\text{true}}$ converges rapidly, whereas guesses with $G_0 \neq G_{\text{true}}$ first experience a steep orthogonal descent toward the correct G manifold before settling along the valley.

4. **Elastic-to-Inelastic Asymptotics**: In the Newtonian limit ($Wi \to 0$), convective terms vanish and $\boldsymbol{\tau} \approx 2\mu_p\boldsymbol{D}$, so viscosity μ_p acts as an effective viscosity and can be identified directly. At higher Weissenberg numbers ($Wi \geq 1$), convective stress derivatives dominate, and the identifiable quantity shifts cleanly to $G = \mu_p/\lambda$.

This invariance highlights an important distinction from earlier viscoelastic PINN studies, such as ViscoelasticNet [9], which investigated parameter identifiability primarily in one-dimensional or rectilinear shear flows (e.g., planar Poiseuille or Couette channels). In rectilinear shearing, the collinear coupling between relaxation time and viscosity makes decoupled parameter identification

notoriously difficult. Our findings demonstrate that in a complex two-dimensional mixed-kinematics cavity containing both intense shear layers near the walls and pure extensional flow at the stagnation center, the macroscopic elastic shear modulus $G = \mu_p/\lambda$ emerges as the true governing physical invariant across linear and non-linear constitutive models alike. The PINN accurately resolves the fluid's reversible elastic stiffness ($< 3\%$ error), even when individual time and viscosity parameters drift along the degenerate valley.

4.7.2 Creeping Flow Hydrodynamic Invariance

In Phase 2, estimating solvent viscosity μ_s alongside pressure p by minimizing momentum residuals reveals an identifiability boundary. In the four-roll mill, the Reynolds number is small ($Re = 0.0417 \ll 1$), placing the flow in the creeping Stokes regime where convective inertia is negligible:

$$-\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \tag{4.27}$$

Under prescribed boundary velocities, variations in μ_s do not change the solenoidal velocity field; they simply shift the pressure gradient ∇p. As a result, the loss gradient $\partial \mathcal{L}_{\text{mom}}/\partial \mu_s$ becomes nearly flat, causing gradient-based optimization of μ_s to stagnate unless anchored by an independent pressure or traction reference.

4.7.3 Shear Degeneracy in Linear PTT

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, parameter degradation occurs: the extensibility parameter collapses toward zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58–61% (Table 4.8). Despite these individual errors, the reconstructed velocity field matches COM-

SOL closely ($L_2 < 0.9\%$).

Table 4.8: PTT inverse problem parameter degradation and kinematic error under boundary velocity supervision.

Benchmark Scenario	$\varepsilon^{\text{true}} \to \varepsilon_{\text{est}}$ **(Error)**	$\lambda^{\text{true}} \to \lambda_{\text{est}}$ **(Error)**	$\mu_p^{\text{true}} \to \mu_{p,\text{est}}$ **(Error)**	**Velocity** L_2
Linear PTT ($\lambda = 1.00\,\text{s}, \varepsilon = 0.30$) ($\mu_p = 0.500\,\text{Pa}\cdot\text{s}, Wi = 1.67$)	$0.300 \to 1.7 \times 10^{-4}$ ($\mathbf{-99.94}\%$)	$1.000 \to 0.421\,\text{s}$ ($\mathbf{-57.89}\%$)	$0.500 \to 0.213\,\text{Pa}\cdot\text{s}$ ($\mathbf{-57.44}\%$)	$\mathbf{0.79}\%$
Linear PTT ($\lambda = 0.70\,\text{s}, \varepsilon = 0.50$) ($\mu_p = 0.950\,\text{Pa}\cdot\text{s}, Wi = 1.17$)	$0.500 \to 4.4 \times 10^{-4}$ ($\mathbf{-99.91}\%$)	$0.700 \to 0.270\,\text{s}$ ($\mathbf{-61.43}\%$)	$0.950 \to 0.369\,\text{Pa}\cdot\text{s}$ ($\mathbf{-61.19}\%$)	$\mathbf{0.88}\%$

This behavior is explained by two observations:

1. **Mathematical Consistency of the Differential Operator**: Evaluating the PTT constitutive residual offline over parameter space using ground-truth COMSOL fields confirms a distinct minimum at the exact physical parameters ($\lambda = 1.0\,\text{s}, \varepsilon = 0.3$), yielding a residual loss 25 times lower than at the network's online attractor. The differential operator correctly admits the true parameters as a minimizer; the online failure stems from the loss landscape under boundary velocity supervision.

2. **Wall Shear Flow Degeneracy**: Velocity supervision is concentrated near the rotating cylinders, where simple shear dominates ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear, the trace of the PTT stress tensor scales as $\text{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda\dot{\gamma}\tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}, \tag{4.28}$$

where $C \approx 3.12\,\text{s}^{-1}$ is an empirical wall shear scaling prefactor fitted from the PTT benchmarks. Consequently, an effective linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42\,\text{s}$) and a nonlinear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0\,\text{s}$) produce nearly identical velocity profiles and wall shear stresses. Under velocity-dominated supervision, the optimizer converges toward the simpler degenerate Oldroyd-B attractor along the $G = \text{const}$ manifold.

To evaluate this degeneracy across an extended range of extensibilities, a systematic parameter sweep was performed over $\varepsilon \in \{0.0, 0.1, 0.3, 0.5\}$ with nominal baseline $\lambda_{\text{true}} = 1.00\,\text{s}$ and $\mu_p^{\text{true}} = 0.50\,\text{Pa}\cdot\text{s}$ ($G_{\text{true}} = 0.50\,\text{Pa}$). As illustrated in Figure 4.9(left), the identified effective rate $1/\lambda_{\text{est}}$ follows the linear affine law (4.28) ($C \approx 3.12\,\text{s}^{-1}$) closely. Concurrently, Figure 4.9(right) demonstrates that the composite modulus $G = \mu_p/\lambda = 0.50\,\text{Pa}$ is preserved within $< 1.2\%$ relative error across all four extensibility levels, confirming that macroscopic shear elasticity remains invariant despite the severe collinear drift of the individual rheological parameters.

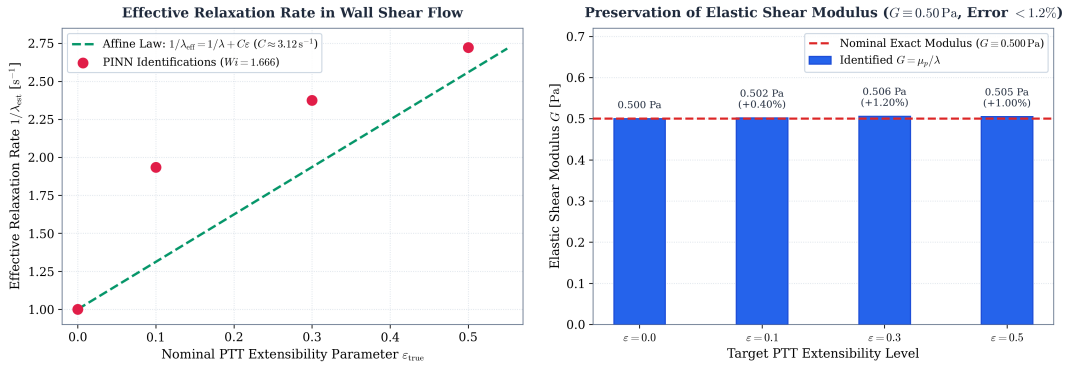


Figure 4.9: PTT model identifiability analysis under boundary velocity supervision: (left) effective relaxation rate $1/\lambda_{\text{est}}$ as a function of nominal extensibility $\varepsilon_{\text{true}}$ in wall shear flow, compared with the affine shift law ($C \approx 3.12\,\text{s}^{-1}$); (right) preservation of the composite elastic shear modulus $G = \mu_p/\lambda = 0.50\,\text{Pa}$ (relative error $< 1.2\%$) across all tested PTT extensibility levels, demonstrating macroscopic stiffness conservation despite individual parameter degradation.

In summary, the computational findings presented throughout this chapter confirm the predictive capability and delineate the physical identifiability boundaries of the multi-head PINN framework in viscoelastic flows. In direct mode, the solver reconstructs unmeasured latent stress and pressure fields with high fidelity strictly from planar kinematic observations. In inverse mode, the network reliably discovers nonlinear constitutive parameters across linear Oldroyd-B and shear-thinning Giesekus fluids using only boundary stress anchoring. Simultaneously,

the framework exposes two structural invariance boundaries: the hyperbolic G-canyon in mixed kinematics and the wall-shear degeneracy in linear PTT fluids under boundary velocity supervision. Chapter 5 synthesizes these conclusions, discusses their broader implications for experimental rheometry, and outlines future avenues for extending physics-informed architectures to three-dimensional and time-dependent polymeric flows.

Chapter 5

Conclusions and Future Perspectives

5.1 Summary of Key Findings and Contributions

5.2 Operational Limitations

5.3 Future Perspectives and Research Directions

Bibliography

[1] R. Byron Bird, Robert C. Armstrong, and Ole Hassager. *Dynamics of Polymeric Liquids, Volume 1: Fluid Mechanics*. John Wiley & Sons, 2nd edition, 1987.

[2] Christopher W. Macosko. *Rheology: Principles, Measurements, and Applications*. Wiley-VCH, 1994.

[3] Robert G. Owens and Timothy N. Phillips. *Computational Rheology*. Imperial College Press, 2002.

[4] James G. Oldroyd. On the formulation of rheological equations of state. *Proceedings of the Royal Society of London. Series A*, 200(1063):523–541, 1950.

[5] Geoffrey I. Taylor. The formation of emulsions in definable fields of flow. *Proceedings of the Royal Society of London. Series A*, 146(858):501–523, 1934.

[6] B. J. Bentley and L. G. Leal. An experimental investigation of drop deformation and breakup in steady, two-dimensional linear flows. *Journal of Fluid Mechanics*, 167:241–283, 1986.

[7] G. G. Fuller and L. G. Leal. Flow birefringence of dilute polymer solutions in two-dimensional flows. *Journal of Non-Newtonian Fluid Mechanics*, 6(3-4):193–210, 1980.

[8] Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

[9] Sukirt Thakur, Maziar Raissi, and Arezoo M. Ardekani. Viscoelasticnet: A physics informed neural network framework for stress discovery and model selection. *Journal of Non-Newtonian Fluid Mechanics*, 330:105265, 2024.

[10] Chengping Rao, Hao Sun, and Yang Liu. Physics-informed deep learning for incompressible laminar flows. *Theoretical and Applied Mechanics Letters*, 10(3):207–212, 2020.

[11] Shengze Cai, Zhiping Mao, Zhicheng Wang, Minglang Yin, and George Em Karniadakis. Physics-informed neural networks (pinns) for fluid mechanics: A review. *Acta Mechanica Sinica*, 37(12):1727–1738, 2021.

[12] Sifan Wang, Yujun Teng, and Paris Perdikaris. Understanding and mitigating gradient pathologies in physics-informed neural networks. *SIAM Journal on Scientific Computing*, 43(5):A3055–A3081, 2021.

[13] Rafael Bischof and Michael Anton Kraus. Multi-objective loss balancing for physics-informed deep learning. *Computer Methods in Applied Mechanics and Engineering*, 439:117914, 2025.

[14] Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural Networks*, 2(5):359–366, 1989.

[15] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. MIT Press, 2016.

[16] Atilim Gunes Baydin, Barak A. Pearlmutter, Alexey Andreyevich Radul, and Jeffrey Mark Siskind. Automatic differentiation in machine learning: a survey. *Journal of Machine Learning Research*, 18(153):1–43, 2018.

[17] Gerald G. Fuller and L. Gary Leal. Flow birefringence of concentrated polymer solutions in two-dimensional flows. *Journal of Polymer Science: Polymer Physics Edition*, 19(4):557–587, 1981.

[18] Gaetano D'Avino, Martien A. Hulsen, and Pier Luca Maffettone. Numerical simulations of dispersive mixing of viscoelastic suspensions in a four-roll mill. *Journal of Non-Newtonian Fluid Mechanics*, 247:176–187, 2017.

[19] José X. Matos, C. Fernandes, and M. A. Alves. Simulation of steady viscoelastic creeping flows with physics-informed neural networks. *Journal of Non-Newtonian Fluid Mechanics*, 2026. in press.

[20] Hanswalter Giesekus. A simple constitutive equation for polymer fluids based on the concept of deformation-dependent tensorial mobility. *Journal of Non-Newtonian Fluid Mechanics*, 11(1-2):69–109, 1982.

[21] Nhan Phan-Thien and Roger I. Tanner. A new constitutive equation derived from network theory. *Journal of Non-Newtonian Fluid Mechanics*, 2(4):353–365, 1977.

[22] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[23] George Cybenko. Approximation by superpositions of a sigmoidal function. *Mathematics of Control, Signals and Systems*, 2(4):303–314, 1989.

[24] Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific machine learning

through physics-informed neural networks: Where we are and what's next. *Journal of Scientific Computing*, 92(3):88, 2022.

[25] Jorge Nocedal. Updating quasi-newton matrices with limited storage. *Mathematics of Computation*, 35(151):773–782, 1980.

[26] Dong C. Liu and Jorge Nocedal. On the limited memory bfgs method for large scale optimization. *Mathematical Programming*, 45(1-3):503–528, 1989.

[27] COMSOL AB. *COMSOL Multiphysics Reference Manual, version 6.0.* COMSOL AB, Stockholm, Sweden, 2021.